AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

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Abstract

Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most existing workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repair cycle computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the repair. This poses a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reducing retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, and avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with the semantic attributes that define the failure slice. To address this limitation, we propose **AttriSelect**, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data coverage, and soft class balance. We provide theoretical analysis and evaluate **AttriSelect** across image classification, pose estimation, and object detection. Across tasks and repair budgets, **AttriSelect** gives the strongest repair results in most settings, showing that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

1 Introduction

Computer vision models can perform well on average while failing on rare semantic subpopulations [4, 15, 7]. For example, a model may recognize people accurately overall but fail specifically on people wearing brown clothes in rainy scenes. Such failures are not fully characterized by the class label alone; instead, they arise from combinations of semantic attributes whose support in the training data may be limited [6, 11]. In deployed systems, the challenge is therefore not only to identify these failure slices, but also to repair them efficiently without repeatedly retraining on the entire dataset.

To better identify and characterize these failures, recent work has developed debugging methods that make failure discovery more interpretable. 3DB [24] builds parameterized simulation spaces for systematic model testing, while HiBug [7] leverages vision–language models to describe failure slices through shared visual attributes. Related approaches such as AdaVision [13], DCD [33], and HiBug2 [8] identify underperforming subpopulations and support retrieval-style interventions. In parallel, generation-based methods modify the training distribution to address model weaknesses. DiGA [42] edits spurious attributes for fair facial-attribute classification, whereas HiBug [7] and

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SafeFix [40] synthesize repair data for diagnosed failures. SafeFix further uses controlled image editing and VLM-based filtering to preserve labels and unrelated attributes.

These pipelines address two parts of the repair workflow: identifying the failing semantic slice and generating targeted repair examples. However, a separate question remains: once targeted images are generated, how should they be combined with the original source data? Retraining on the full source pool can be costly, especially for dense prediction tasks. For example, training Keypoint R-CNN [16] for COCO person keypoint estimation [25] requires approximately eight GPU-days [12]. As new bug slices are discovered across repair iterations, repeatedly retraining with the full source dataset becomes increasingly expensive.

Selecting a compact subset of source data is a natural strategy for reducing the cost of iterative repair, making the problem closely related to active learning (AL). Classical AL methods such as entropy-based uncertainty sampling [32], CoreSet [31], and BADGE [1] select compact training subsets using model uncertainty, feature-space coverage, or gradient embeddings. However, targeted repair introduces challenges beyond standard AL settings. The selector must preserve useful source coverage while allocating enough budget to the diagnosed attribute slice, and it must account for the distribution shift from generated repair images. As a result, a generic uncertainty- or coverage-based score may select informative images while still omitting rare repair-relevant attribute values, whereas an attribute-greedy policy may over-select values already supplied by the generated data.

We study this remaining step as a repair-time source-selection problem. The input is not an unlabeled pool, but a diagnosed failure slice, a generated repair set, and an annotated source pool. This changes the selection objective: the subset should preserve useful source coverage, include real examples with rare repair-relevant attributes, and avoid spending budget on attribute values already supplied by the generated images. We propose **AttriSelect** for this setting. Rather than representing source images only through feature embeddings or gradient statistics, **AttriSelect** treats each image as a structured attribute record and scores it using class-conditional rarity, generated-data frequency control, and soft class balancing. In this role, **AttriSelect** is a focused selection component for existing repair pipelines: it does not replace failure-slice discovery or repair-image generation, but decides which original training examples should accompany the generated data during retraining.

Our contributions are:

1. We formulate targeted vision repair as a repair-time source-selection problem: given a diagnosed failure slice, generated repair data, and an annotated source pool, the goal is to select a compact source subset for retraining.
2. We propose **AttriSelect**, an attribute-driven source selector that combines class-conditional rarity, generated-data frequency control, and soft class balancing to prioritize repair-relevant source examples while avoiding redundancy and class collapse.
3. We evaluate **AttriSelect** across image classification, pose estimation, and object detection, comparing against uncertainty-based, coverage-based, gradient-embedding, detector-oriented, and random or stratified baselines under matched repair budgets. Quantitatively, **AttriSelect** gives the best performance in almost all active selection settings; with only half of the source data, it already exceeds the full-source repair reference in multiple tasks, while selecting subsets at least $39\times$ faster than the fastest non-attribute active-learning baseline after attribute records are available.

2 Related Work

Attribute-Based Debugging and Repair. Attribute-based debugging methods expose model failures as human-interpretable semantic slices. AdaVision [13] and DCD [33] identify underperforming subpopulations and use retrieval-style interventions, which can require external data and manual validation. HiBug [7] uses vision–language models to discover failure slices through shared visual attributes, and HiBug2 [8] improves the efficiency and coverage of this process. SafeFix [40] shifts from retrieval to controlled generation by editing real training instances and filtering generated images with a vision-language model (VLM). These works focus on finding bug slices or constructing repair examples. We study the remaining selection problem: given source data and targeted generated data, select a compact retraining subset that preserves coverage while emphasizing repair-relevant attributes.

Active Learning. Active learning reduces annotation or retraining cost by selecting informative subsets under a budget. Standard approaches include entropy-based uncertainty sampling [32], coreset selection [31], and gradient-embedding methods such as BADGE [1]. Related data subset-selection methods use submodular, generalization-based, gradient-matching, and distribution-shift objectives [38, 20, 19, 22, 18]. Object-detection and label-budget-robust variants combine these signals, including PPAL [41] and Uncertainty Herding [2]. Targeted active learning has also considered rare classes and slices for object detection [23]. Our setting differs from standard AL because the goal is not generic sample efficiency, but selecting source data that complements a targeted repair set and corrects an attribute-defined failure slice.

3 Background

Attribute-Based Failure Slices. A failure slice is a group of examples that share a semantic description and have poor model performance. For instance, a pose model may fail on people with brown clothing in rainy scenes, even if its average performance is high. Attribute-based debugging methods make this description explicit by attaching semantic attributes to each image.

Formally, let $f_\theta : \mathcal{X} \to \mathcal{Y}$ be the vision model and let each image $x \in \mathcal{X}$ have attributes $\mathcal{A} = \{a_1, \ldots, a_m\}$. Each attribute $a_j : \mathcal{X} \to \mathcal{V}_j$ maps an image to a discrete value, such as a color, weather condition, texture, or viewpoint. A slice defined by several attribute values is $T = \{x \mid a_{j_1}(x) = v_{j_1}, \ldots, a_{j_k}(x) = v_{j_k}\}$. We focus on slices that are both rare in the training set $\mathcal{D}_{\text{train}}$ and weak on the validation set $\mathcal{D}_{\text{val}}$. In other words, the slice has low support and its validation accuracy is substantially below the overall validation accuracy.

Source Data and Generated Repair Data. After a slice has been diagnosed, repair uses two kinds of images. The first is the original source pool, denoted $\mathcal{D}_{\text{src}} \subseteq \mathcal{D}_{\text{train}}$. These images preserve the broad class and context coverage learned by the model. The second is a generated repair pool, denoted $\mathcal{D}_{\text{gen}}$. These images are produced for the diagnosed slice by pipelines such as HiBug and SafeFix [7, 40]; they are intended to satisfy the target attributes while preserving the original label. Appendix I tests the selection result with both HiBug-generated repair data and alternative image generators.

Data Selection for Repair. The repair question is which source images should be kept when retraining with the generated images. Keeping all of $\mathcal{D}_{\text{src}}$ is expensive, but keeping an arbitrary small subset can remove useful coverage. We write the retraining set as $\mathcal{D}_{\text{repair}} = \mathcal{D}_{\text{sel}}^{\text{src}} \cup \mathcal{D}_{\text{sel}}^{\text{gen}}$, where $\mathcal{D}_{\text{sel}}^{\text{src}} \subseteq \mathcal{D}_{\text{src}}$ is the selected source subset and $\mathcal{D}_{\text{sel}}^{\text{gen}} \subseteq \mathcal{D}_{\text{gen}}$ is the generated subset used for repair. In our experiments, $\mathcal{D}_{\text{sel}}^{\text{gen}}$ is fixed to 1,000 generated images, so the main decision is the choice of $\mathcal{D}_{\text{sel}}^{\text{src}}$. A useful selector should keep class coverage, add real examples that complement the generated data, and avoid spending source budget on attribute values already abundant in $\mathcal{D}_{\text{gen}}$.

4 Methodology

4.1 AttriSelect Framework

AttriSelect is designed as the selection step inside an existing targeted repair pipeline. Figure 1 summarizes this workflow. The upstream workflow can follow prior model-repair systems such as HiBug and SafeFix [7, 40]. In the experiments reported here, the task-level attribute set $\mathcal{A}$ is fixed before selection, and Qwen3-VL [3] assigns an attribute value $v_{i,a}$ to each training or validation image. Appendix D gives the task-specific attribute vocabularies and assignment protocol. Appendix H reports a human audit and attribute-noise sensitivity, while Appendix J evaluates incomplete target-field coverage. Using the SafeFix-style ImageNet10 classification setting as a running example, a ResNet-18 classifier [17] is then trained to a checkpoint θ_t, after which the repair pipeline analyzes the validation behavior of f_{θ_t}. An attribute slice T_{bug} is marked as a rare failure case if it is underrepresented in training and underperforms on validation:

$$|T_{\text{bug}} \cap \mathcal{D}_{\text{train}}| < \rho |\mathcal{D}_{\text{train}}|, \qquad \text{Acc}(T_{\text{bug}}) < \text{Acc}(\mathcal{D}_{\text{val}}), \tag{1}$$

where ρ is a user-specified rarity threshold. For example, the ImageNet10 pipeline may diagnose a slice whose attribute is *texture* and whose value is *fabric*. A conditional diffusion model, such as ControlNet [43], then generates repair images of objects with fabric texture while preserving the class label and other attributes.

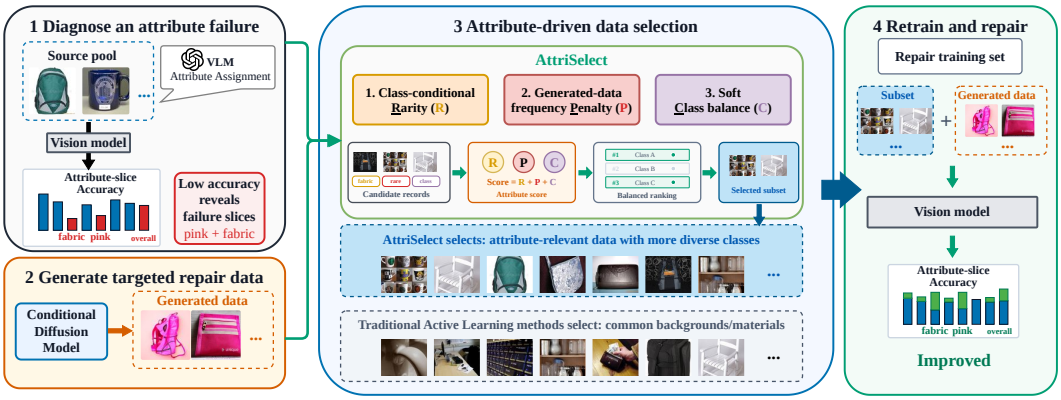


Figure 1: **AttriSelect** framework. Given a diagnosed failure slice and generated images, **AttriSelect** scores source examples with class-conditional rarity, generated-data frequency penalty, and soft class balance, then returns an attribute-driven subset for retraining with the generated repair set.

AttriSelect starts after this upstream diagnosis and generation stage. It takes as input the diagnosed slice, the original source pool with VLM-assigned attributes, and the generated repair images for that slice. The output is a compact source subset $\mathcal{D}_{\text{sel}}^{\text{src}}$, which is combined with the generated images and used to continue training or fine-tune the checkpoint θ_t. Thus **AttriSelect** does not replace VLM-based attribute proposal, failure-slice discovery, or conditional image generation. It decides which original training examples should accompany the generated repair data so that retraining preserves source coverage while allocating more budget to real examples that complement the diagnosed repair slice. Appendix K separately evaluates the selection rule without a repair checkpoint, diagnosed slice, or generated data, including one-shot and iterative active-learning settings with multiple attribute sources.

4.2 Attribute-Driven Active Learning

Given $\mathcal{D}_{\text{src}}$ and $\mathcal{D}_{\text{gen}}$ for a discovered slice, the goal is to choose a small source subset that still supports the repair. Active learning is a natural tool for this problem: entropy-based uncertainty sampling selects examples with high predictive entropy [32], while CoreSet [31], BADGE [1], PPAL [41], and Uncertainty Herding [2] add coverage, gradient diversity, or detector-side difficulty signals.

For targeted repair, however, the subset must answer a semantic question: which real images best complement the generated examples for the diagnosed failure slice? A subset may provide good coverage of the source distribution in terms of class labels or feature space while still containing very few examples from the rare failure slice itself. Likewise, entropy-based sampling can miss the slice entirely if the model assigns higher uncertainty to unrelated images. Proposition 1 formalizes this base-rate mismatch.

A second challenge arises because generated repair images alter the attribute distribution seen during retraining. If the generated pool already contains many examples with a target attribute, the source selector should not continue allocating most of its remaining budget to redundant examples of the same attribute. For instance, if generated repair images already emphasize rainy weather, a purely attribute-greedy selector may over-select common rainy street scenes while failing to retain rainy examples from underrepresented classes, unusual viewpoints, or occluded settings. As a result, the repair process can become biased toward the easiest contexts in which the target attribute appears.

AttriSelect uses a three-part scoring function to avoid these failure modes. First, the **class-conditional rarity** term prioritizes attribute values that are rare within their class and therefore more likely to be repair-relevant. Second, the **generated-data frequency penalty** downweights attribute values that are already well represented by the generated repair images, reducing redundancy between synthetic and source data. Third, the **soft class-balance** term encourages coverage across classes by increasing the priority of classes that have received insufficient source budget. Together, these components steer selection toward repair-relevant attributes while preventing the subset from collapsing onto repeated synthetic attributes or a small number of easy classes.

Class-conditional rarity. For a source image x_i with class label y_i, let $v_{i,a}$ be the value of attribute a on that image. The rarity term gives higher weight to attribute values that occur fewer times within the same class:

$$R_{i,a} = \left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} \tag{2}$$

where $n_a(v_{i,a} \mid y_i)$ is the number of source examples in class y_i whose attribute a takes value $v_{i,a}$, $\varepsilon > 0$ is a numerical stability constant, and $\alpha \geq 0$ controls how strongly the selector prefers rare attribute values.

This term remains important even after adding generated repair images. Although generated images can make a target attribute less rare in the combined repair pool, they do not make real source examples with that attribute redundant, since the two data sources may differ in image statistics, contextual diversity, learned representations, and other unannotated visual factors. Because repair problems often originate from rare-case failures, class-conditional rarity helps retain real source examples containing rare repair-relevant attributes in the candidate ranking. The role of the generated-data frequency penalty is complementary: rather than promoting rarity, it prevents this preference from repeatedly selecting attribute values that are already heavily represented in the generated repair data.

Generated-data frequency penalty. Rarity alone can over-select attribute values that are already well covered by the generated repair pool. In the rainy-weather example above, if the generated data already contains many rainy images, the selector should not spend most of its remaining source budget on additional easy rainy cases from similar contexts. Instead, it should retain useful real-source rainy examples while still leaving room for complementary classes, viewpoints, and occlusion patterns. To capture this trade-off, we measure how common each attribute value becomes after combining source and generated data as $\hat{n}_a(v) = n_a^{\mathrm{src}}(v) + \tilde{n}_a^{\mathrm{gen}}(v)$, where $n_a^{\mathrm{src}}(v)$ denotes the source count and $\tilde{n}_a^{\mathrm{gen}}(v)$ denotes the generated count used by the selector after any scaling of the generated pool. Attribute values with larger effective counts receive lower scores:

$$P_{i,a} = -\lambda \cdot \frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \tag{3}$$

where $\lambda \geq 0$ controls how strongly the selector penalizes already over-represented attribute values.

We then define the **base source score** $\bar{s}_i$ by summing the rarity and generated-data frequency terms over the repair attributes:

$$\bar{s}_i = \sum_{a \in \mathcal{A}} \Big(R_{i,a} + P_{i,a} \Big). \tag{4}$$

Each component addresses a different failure mode in repair-time selection. Setting $\alpha = 0$ removes the preference for rare semantic configurations within a class, reducing the selector to a frequency-driven policy. Setting $\lambda = 0$ removes the redundancy penalty, allowing the selected subset to repeatedly choose attribute values already well represented in $\mathcal{D}_{\mathrm{gen}}$. Without class-aware re-ranking, high-scoring examples may further concentrate within a small subset of classes. As a result, the score is not intended to produce a neutral representative subset; instead, it explicitly allocates source budget toward semantic coverage that is most useful for the diagnosed repair problem. Appendix F compares the full score with a Generated-Complement control and reports the resulting class–attribute coverage, attribute diversity, and class balance of the selected subsets.

Soft class balance. Attribute scores are computed per image, so they can still over-select classes that happen to contain many high-scoring attributes. For example, in a fabric-texture repair, many high-scoring images may come from classes where fabric is visually common, while guitar images may contain the texture only in straps or carrying cases. We correct this with a soft class-balance term. Let n_{select} be the source-data budget and let N_{cls} be the number of classes, so the per-class target is $t = n_{\mathrm{select}}/N_{\mathrm{cls}}$. During greedy selection step k, a candidate from class c receives the adjusted score

$$\tilde{s}_i^{(k)} = \bar{s}_i + C_c^{(k)}, \qquad C_c^{(k)} = w \cdot \frac{t - n_c^{(k)}}{t}, \tag{5}$$

where $n_c^{(k)}$ is the number of already selected source samples from class c, and $w \geq 0$ controls the strength of soft class balancing. Classes below their target allocation receive a positive correction,

while classes above their target receive a negative one. As w increases, the selector behaves more like stratified sampling, with the limiting case $w = \infty$ corresponding to round-robin class quotas. This mechanism preserves the attribute-based ranking from Eq. 4 within each class while preventing a small subset of classes from dominating the repair set.

Selection is performed greedily. We first compute the base score $\bar{s}_i$ for every source candidate and initialize both the selected set and the per-class counts to zero. At each iteration, every remaining candidate receives the adjusted score $\tilde{s}_i^{(k)}$ from Eq. 5. The highest-scoring candidate is then added to $\mathcal{D}_{\text{sel}}^{\text{src}}$, its class count is updated, and the soft class-balance corrections are recomputed. The process repeats until n_{select} source examples have been selected. The generated repair set remains fixed throughout this procedure and influences selection only through the frequency counts in Eq. 3. As a result, all methods are evaluated under the same generated-data budget. Appendix C gives the full pseudocode, and Appendix M reports the score hyperparameters used in the experiments.

4.3 Theoretical Analysis

Useful repair subsets should contain rare, underperforming semantic configurations that complement the generated data. Let $N = |\mathcal{D}_{\text{src}}|$ be the source-pool size and $m = \lfloor rN \rfloor$ be the budget for $0 < r < 1$. Let $T \subseteq \mathcal{D}_{\text{src}}$ denote the target failure slice, and write $T_c = T \cap \{x_i : y_i = c\}$ and $N_c = |\{x_i : y_i = c\}|$ for class-conditional support.

Proposition 1 (Attribute-agnostic AL requires target alignment)**.** *Consider any AL selector that, within class c, chooses B_c source examples using only non-attribute information h_i, such as uncertainty, logits, embeddings, losses, detector difficulty, coverage, or image distances, and does not use the attribute values $\{v_{i,a} : a \in \mathcal{A}\}$. Let $\mathcal{D}_{\text{AL}}$ denote its selected source subset. If the target-slice indicator $\mathbf{1}[x_i \in T_c]$ is conditionally independent of h_i given class c, with ties and internal randomness independent of the target indicator, then*

$$\mathbb{E}\big[|\mathcal{D}_{\text{AL}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c}. \tag{6}$$

Thus an attribute-agnostic AL method can enrich the target slice only when its non-attribute signal is additionally aligned with that slice.

The expectation in Eq. 6 reflects a base-rate limit: if a slice is only 2% of class c, selecting 100 class-c examples yields only 2 target examples in expectation. **AttriSelect** avoids relying on such indirect alignment by observing the attributes that define the diagnosed slice and prioritizing rare repair-relevant values directly. The generated-data penalty and soft class-balance term keep this preference from repeatedly selecting synthetic-like attributes or collapsing onto a few classes. Coverage-based selectors face a similar issue: if target examples form a compact feature region, one or two representatives may cover that region geometrically, even though repair may need more examples from it.

The proposition is stated at the level of selector information rather than for a specific baseline. Stronger baselines such as PPAL and Uncertainty Herding still depend on uncertainty, difficulty, or coverage being aligned with the discovered attribute slice; when this alignment is weak, allocation to the target slice remains proportional to slice prevalence.

See Appendix L for the **complete proof**, a coverage-based instance, and score-level conditions under which **AttriSelect** prioritizes rare target-slice examples while limiting class imbalance.

5 Experiments

Datasets and Tasks We evaluate **AttriSelect** on three vision tasks: image classification, pose estimation, and object detection. In all settings, the goal is to test whether attribute-aware source-data selection can choose a subset that improves the model performance and fixes bugs.

For image classification, we follow SafeFix [40]: a ResNet-18 classifier on an ImageNet subset [9] with 10 classes, 1,300 images per class, and an 8:1:1 train/validation/test split, denoted ImageNet10.

We also evaluate CelebA [26] lipstick-wearing classification with a ResNet-18-based binary classifier and 80,000/10,000/10,000 train/validation/test images. For pose estimation, we follow HiBug2 [8] on COCO person keypoints [25], using Keypoint R-CNN [16] with a ResNet-50-FPN backbone [17], 20,000 source training images, and 1,173-image held-out validation and test splits, evaluated by COCO keypoint AP. For object detection, we use KITTI [14] Car and Pedestrian instances converted to YOLO format with 5,000/1,240/1,241 train/validation/test images, and train YOLOv8n. Implementation details are given in Appendix B.

Metrics Let M denote the task metric: test accuracy for image classification, COCO keypoint AP for pose estimation, and mAP50–95 for object detection. We focus on targeted improvements for failure cases, which are critical for fairness and safety, while also tracking overall task performance. Following standard practice in machine learning classification evaluation [27], we report Relative Error Reduction (RER),

$$\mathrm{RER}(\%) = 100 \cdot \frac{M_{\mathrm{after}} - M_{\mathrm{before}}}{1 - M_{\mathrm{before}}}, \tag{7}$$

where M_{before} and M_{after} are the test metrics before and after repair. In the main RER tables, M_{before} is 0.7173 for ImageNet10, 0.9057 for CelebA, 0.5902 for COCO, and 0.5626 for KITTI. For accuracy, RER is the percentage of the original error rate removed by repair; for AP and mAP, it is the normalized improvement toward the maximum score, reported in percent.

Baselines We compare **AttriSelect** with seven source-selection baselines. **RANDOM** samples images uniformly at random. **STRATIFIED** preserves class proportions during sampling [28]. **ENTROPY** is the entropy-based uncertainty baseline; it selects examples with the largest predictive entropy or task-specific confidence-entropy proxy [32]. **CORESET** uses greedy coverage in embedding space [31]. **BADGE** selects gradient-embedding batches that combine uncertainty and diversity [1]. **PPAL** is a plug-and-play active-learning baseline for object detection [41]. **UHERDING** denotes Uncertainty Herding, which combines uncertainty with coverage across label budgets [2].

The seven generic baselines do not receive the repair attributes used by **AttriSelect**. Appendix F therefore adds matched-information baselines that use the same target fields through direct filtering, stratification, entropy ranking, or generated-set complementarity.

5.1 Main Results

Table 1 compares **AttriSelect** with generic source-selection baselines using RER. All main-result entries are averaged over five random seeds. The original task metrics and variance over five seeds are reported in Appendix E.

Analysis. **AttriSelect** gives the best RER in almost all active source-selection settings, excluding the ratio-1.0 full-data references where all selectors are identical. The gap is largest at small budgets, where the selected source subset is highly constrained, and it generally remains at larger budgets. KITTI is the only near-exception: PPAL is slightly higher at ratio 0.7 by 0.05 RER points, while **AttriSelect** remains best at the other four KITTI ratios. Appendix I reports the same selector ordering under an alternative repair pipeline and alternative generators, and Appendix J shows that the ImageNet10 ordering persists across longer repair schedules.

Subset selection can also outperform the full-data setting, especially on ImageNet10, CelebA, and COCO. This suggests that the full candidate pool contains noisy, redundant, or target-irrelevant samples, and that selection improves the effective training distribution. In this sense, **AttriSelect** is not only a way to save data or training cost; it also improves data quality and target-slice alignment during repair. Appendix F shows that **AttriSelect** remains strongest at the more constrained source ratios 0.02 and 0.05 and quantifies the class–attribute coverage and diversity of the selected subsets. The t-SNE visualizations in Appendix O show that **AttriSelect** does not simply trace the same regions as generic AL baselines, but selects repair-attribute-related source examples across varied visual contexts.

KITTI behaves differently at the smallest source budget. At ratio 0.1, all methods have negative RER. A likely reason is that KITTI has only $5{,}000$ training images, so the selected source subset contains just 500 real images, unlike ImageNet10, COCO, and CelebA where the same ratio still leaves thousands to tens of thousands of source examples. This negative value should therefore be read as a low-data failure mode: for small detection datasets, an overly small repair subset may

Table 1: Main repair results across tasks, averaged over five random seeds. Metric: RER (%).

Dataset	Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
ImageNet10	0.1	3.40	3.47	2.87	3.40	5.31	5.59	3.96	**6.93**
	0.3	4.78	6.93	4.21	6.93	8.84	10.47	7.75	**11.28**
	0.5	7.32	9.66	5.59	8.03	10.47	8.38	8.84	**11.85**
	0.7	8.84	6.12	6.40	7.75	10.75	10.47	8.84	**12.10**
	0.9	10.22	10.47	10.22	6.12	9.37	11.57	11.85	**13.76**
	1.0	8.03	8.03	8.03	8.03	8.03	8.03	8.03	8.03
CelebA	0.1	4.88	2.44	7.10	-6.04	3.18	-1.06	5.94	**8.48**
	0.3	6.04	5.30	8.70	7.95	1.91	3.50	4.03	**8.91**
	0.5	6.04	7.95	7.64	8.17	2.86	2.23	9.97	**13.79**
	0.7	2.86	8.27	10.18	6.57	8.06	5.09	6.04	**12.09**
	0.9	3.18	4.56	8.27	6.04	6.04	4.56	6.57	**14.74**
	1.0	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04
COCO	0.1	3.56	5.08	5.61	-0.61	2.03	2.71	1.17	**6.86**
	0.3	3.37	8.32	7.64	8.64	6.12	4.32	6.42	**9.86**
	0.5	7.49	8.00	9.27	6.47	7.42	6.12	6.83	**10.13**
	0.7	6.08	9.69	9.54	9.10	8.64	6.17	7.83	**10.18**
	0.9	4.61	9.49	6.54	7.32	7.59	7.05	9.74	**11.05**
	1.0	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71
KITTI	0.1	-9.65	-7.27	-7.52	-8.89	-6.95	-9.24	-8.32	**-5.81**
	0.3	-0.48	-1.23	0.23	-0.32	-0.05	-1.21	-1.33	**0.46**
	0.5	2.13	1.23	3.20	3.04	2.29	1.12	2.17	**3.36**
	0.7	3.20	3.06	3.34	3.34	2.08	3.25	**3.57**	3.52
	0.9	4.37	4.66	4.66	3.54	4.21	4.85	3.89	**5.12**
	1.0	5.30	5.30	5.30	5.30	5.30	5.30	5.30	5.30

Table 2: Source-subset selection time. Units are seconds (s), minutes (m), and hours (h). Speedup is measured against the fastest non-attribute active-learning baseline in the same row.

Dataset	Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours / speedup**
COCO	0.1	0s	0s	863s	5m	21m	86s	67s	**0.29s / 231**×
COCO	0.3	0s	0s	865s	15m	1h	93s	85s	**0.30s / 283**×
COCO	0.5	0s	0s	865s	25m	2h	94s	90s	**0.32s / 281**×
CelebA	0.1	0s	0s	49s	45m	2h	63s	46s	**0.42s / 110**×
CelebA	0.3	0s	0s	51s	3h	5h	65s	51s	**0.51s / 100**×
CelebA	0.5	0s	0s	54s	5h	9h	61s	61s	**1.37s / 39**×

not retain enough background, viewpoint, and object-scale variation to improve over the original detector. Under the same generated repair set and source budget, however, **AttriSelect** remains the least damaging option and has the highest RER in this row, indicating the strongest relative source subset under the constrained comparison.

5.2 Computational Efficiency

After attribute records are available, **AttriSelect** is also cheaper than generic active-learning selectors. Its selection cost is dominated by counting VLM-assigned attribute values and sorting candidates, while uncertainty, coverage, and gradient-embedding baselines require model scoring, embedding comparisons, or iterative diversity updates. Table 2 reports source-subset selection time on COCO and CelebA for the three ratios used in the main small-to-medium budget comparison. The source pools contain 20,000 COCO training images and 80,000 CelebA training images. Random and stratified sampling have negligible algorithmic selection cost after excluding data loading and export overhead; we keep them in the table as zero-cost references.

AttriSelect selects each subset in less than two seconds. The fastest non-attribute active-learning baseline takes tens of seconds, and coverage or gradient-embedding methods can take minutes to hours. Even against the fastest nontrivial baseline in each row, **AttriSelect** is at least 39× faster

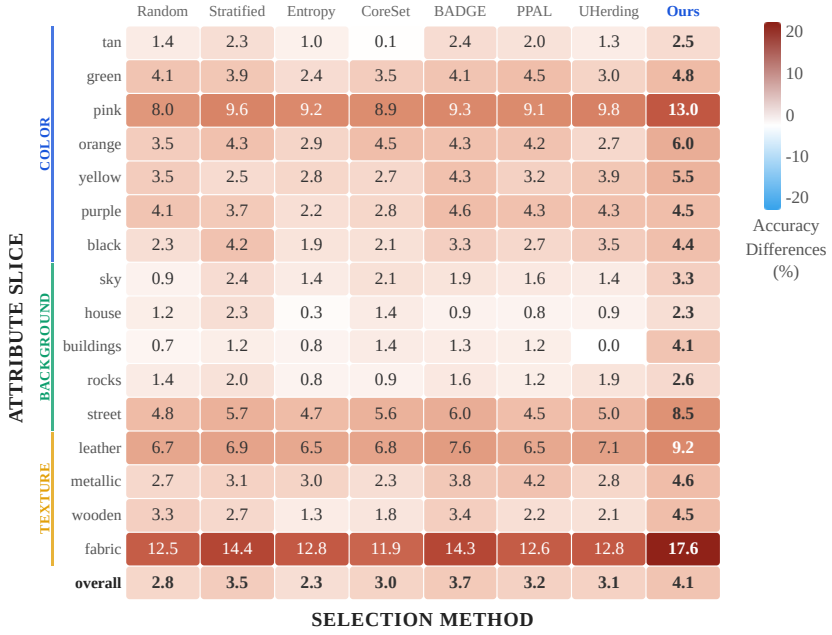


Figure 2: ImageNet10 ratio-0.5 attribute-slice accuracy differences. **AttriSelect** gives larger gains on the targeted attribute families, with the strongest gains on the exact repair values.

on CelebA and more than 230× faster on COCO. These are source-subset selection speedups, not end-to-end pipeline speedups. Appendix B.1 separately reports VLM assignment and first-repair wall-clock time, as well as the lower cost after source metadata are reused.

5.3 Attribute-Slice Accuracy Differences

For each attribute slice, we report the accuracy difference $M_{\text{after}} - M_{\text{before}}$, where M_{before} is measured at the original vision model checkpoint θ_t and M_{after} is measured after retraining. For ImageNet10 classification, M is accuracy. Positive values therefore indicate slice-level improvement after repair. Figure 2 reports these differences for ImageNet10 at source ratio 0.5.

All methods use the same generated repair images and differ only in the selected source subset. The generated data targets pink color and fabric texture. **AttriSelect** gives the largest gains on the targeted attribute families, especially the exact repair values pink and fabric, indicating that the selected source subset improves target-slice alignment rather than only changing overall accuracy. Appendix G provides the COCO pose-estimation heatmap over the full annotated vocabulary. Because these heatmaps show marginal attribute values, Appendix F also reports the full diagnosed conjunction and its strict complement. Appendix J evaluates how the result changes when the slice uses one, two, or three target fields.

5.4 Failure-Slice Repair Counts

We also evaluate repair at the level of discovered failure slices. In the upstream bug-discovery stage for COCO, 12 failure slices are identified. With the proposed framework, **AttriSelect** repairs 10 of these slices already at source ratio 0.1, whereas the strongest active-learning baseline repairs 7 slices at the same ratio. The strongest baseline reaches 10 repaired slices only from source ratio 0.5 onward, while **AttriSelect** repairs 11 slices at ratios 0.5, 0.7, and 0.9.

Table 3: COCO repaired slices out of 12.

Ratio	Best AL	Ours
0.1	7	**10**
0.3	8	**10**
0.5	10	**11**
0.7	10	**11**
0.9	10	**11**

5.5 Ablation Study

We ablate the three score components on ImageNet10 classification repair to test whether the gains come from a single term or from their combination. Table 4 reports test accuracy for all component subsets. The full $R + P + C$ setting combines rarity R, generated-data frequency penalty P, and

Table 4: ImageNet10 classification attribute-score component ablation. Metric: test accuracy.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$\boldsymbol{R+P+C}$
0.1	0.7269	0.7300	0.7262	0.7277	0.7359	0.7308	0.7315	**0.7369**
0.3	0.7308	0.7415	0.7392	0.7392	0.7308	0.7369	0.7315	**0.7492**
0.5	0.7400	0.7415	0.7408	0.7346	0.7369	0.7415	0.7446	**0.7508**
0.7	0.7423	0.7469	0.7431	0.7415	0.7431	0.7469	0.7477	**0.7515**
0.9	0.7462	0.7500	0.7526	0.7477	0.7508	0.7415	0.7423	**0.7562**

soft class balance C. The full $R+P+C$ setting is the most stable choice in this ablation, giving the best result at every ratio. The leave-one-component-out drops from the full setting are nearly identical: removing R drops test accuracy by 0.00942 on average, removing P by 0.00940, and removing C by 0.00940. Thus the ablation supports a complementarity claim rather than a single dominant-component claim on ImageNet10. Single-component variants are weaker on average (R: 0.74198, P: 0.74038, C: 0.73814) than the full setting (0.74892), showing that each component alone is insufficient. A COCO pose-estimation component ablation with a broadly similar trend is reported in Appendix N.

6 Conclusion

We studied targeted vision model repair as a repair-time source-selection problem. In this setting, a failure slice has been diagnosed and targeted repair images have been generated; the remaining question is which source examples should accompany those generated images during retraining. **AttriSelect** addresses this question with an attribute-driven score that combines class-conditional rarity, generated-data frequency control, and soft class balancing. The analysis shows when this score enriches rare target slices relative to attribute-agnostic selectors, and the experiments across classification, pose estimation, and detection indicate that structured attribute metadata provides a useful signal beyond uncertainty, coverage, and gradient-embedding scores. **AttriSelect** is therefore best viewed as one component in a localized repair pipeline, not as a replacement for upstream bug discovery or image generation.

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A Glossary

The notations are summarized in Table 5.

Table 5: Glossary of variables and symbols used in this paper.

Symbol	Description
$\mathcal{X}$	Input image space
$\mathcal{Y}$	Label space
$\mathcal{D}_{\text{src}}$	Source training image dataset
$\mathcal{D}_{\text{gen}}$	Generated repair image pool
$\mathcal{D}_{\text{repair}}$	Retraining dataset composed of selected source and generated images
$\mathcal{D}_{\text{sel}}^{\text{src}}$	Selected subset of the source dataset for retraining
$\mathcal{D}_{\text{sel}}^{\text{gen}}$	Generated subset used for repair
$\mathcal{A}$	Task-level attribute vocabulary proposed by the VLM
$\mathcal{V}_a$	Vocabulary of candidate values for attribute $a \in \mathcal{A}$
$v_{i,a}$	Attribute value assigned to image x_i for attribute a
s_i	Final selection score of source candidate x_i
$\bar{s}_i$	Base selection score (rarity score minus frequency penalty)
$n_a(v \mid c)$	Class-conditional count of attribute value v in class c in the source pool
$\hat{n}_a(v)$	Frequency count of attribute value v in the generated repair pool
$n_c^{(k)}$	Number of selected source samples from class c at selection iteration k
α	Rarity exponent
λ	Frequency penalty weight
w	Soft class-balancing weight
T	Target failure slice diagnosed by the debugging pipeline
T_c	Class-conditional target failure slice
B_c	Number of selected class-c examples (source budget)
N_c	Total support size of class c in the source pool

B Implementation Details

Random seeds. For the main-result tables, we run each setting with five seeds: 0, 42, 1234, 3407, and 2024. Unless stated otherwise, auxiliary analyses outside the main-result tables use seed 42.

Hardware and datasets. Each experimental run is executed on a single NVIDIA H200 GPU; no run uses multi-GPU training. For image classification, we follow SafeFix [40]: a ResNet-18 classifier [17] on an ImageNet subset [9] with 10 classes, 1,300 images per class, and an 8:1:1 train/validation/test split, denoted ImageNet10. The ImageNet10 classes are `backpack`, `barber chair`, `coffee mug`, `desk`, `electric guitar`, `park bench`, `pitcher`, `purse`, `rocking chair`, and `water bottle`. We also evaluate CelebA [26] lipstick-wearing classification with a ResNet-18-based binary classifier and 80,000/10,000/10,000 train/validation/test images. For pose estimation, we follow HiBug2 [8] on COCO person keypoints [25], using Keypoint R-CNN [16] with a ResNet-50-FPN backbone [17], 20,000 source training images, and 1,173-image held-out validation and test splits. For object detection, we use KITTI [14] Car and Pedestrian instances converted to YOLO format with 5,000/1,240/1,241 train/validation/test images, and train YOLOv8n.

Training and repair schedule. For ImageNet10 classification, the diagnosis-stage ResNet-18 model is trained with cross-entropy loss and Adam [21], using learning rate 1×10^{-4}, weight decay 1×10^{-3}, batch size 64, and 30 epochs. During repair, we train with the selected source subset and the generated repair images for 15 epochs using learning rate 1×10^{-5}. For CelebA, COCO, and KITTI, the repair-stage epoch count is set to half of the corresponding diagnosis-stage training schedule, with the same repair-stage learning rate 1×10^{-5}.

Image generation. For the generated repair images used in the main results, we use ControlNet [43] with the Stable Diffusion 1.5 checkpoint, whose backbone follows the latent diffusion architecture [30]. The model is conditioned on soft HED boundaries [39] extracted from source

images and uses 30 DDIM inference steps [34]. The soft boundary condition helps preserve object structure during edits, which is useful for attribute changes such as recoloring, material changes, and weather or style edits.

Attribute proposal and assignment. We use vision-language models (VLMs) in two separate roles. First, a VLM proposes the semantic attribute vocabulary $\mathcal{A}$ for a task. For example, in KITTI object detection, GPT-5 [29] is used to define candidate visual attributes such as color, weather, scale, viewpoint, occlusion, truncation, and motion state. Second, a VLM assigns attribute values to source and generated images. In the experiments reported in this paper, we use Qwen3-VL [3] for this assignment step. For the KITTI object-detection repair, color and weather are the attribute fields tied to the discovered repair slice.

Generation and assignment cost. On one NVIDIA H200 GPU, generating 1,000 T2I repair images with ControlNet takes approximately 20–30 minutes, and assigning repair-relevant attributes to these 1,000 generated images with Qwen3-VL takes approximately 5–10 minutes. The assignment time depends on the number of queried attributes, as reflected by the normalized source-pool measurements below. These shared upstream costs are reported separately from source selection and repair training because every selector uses the same fixed generated set. The source-pool assignment times reported below use the same one-H200 setting.

B.1 End-to-End Cost Accounting

Table 2 isolates source-subset selection after attribute records are available. We additionally measure the one-time source-pool assignment cost and the complete first repair run. T2I generation and generated-image assignment are reported separately because every selector uses the same fixed generated set. Source-pool assignment is charged to **AttriSelect** because it is a method-specific prerequisite when suitable metadata do not already exist.

Table 6: Measured source-pool attribute-assignment cost on one NVIDIA H200 GPU. The normalized time divides the total assignment time by the source-pool size in thousands and the number of assigned attributes.

Dataset	Source images	Attributes	Assignment time	min / 1K / attr.
KITTI	5,000	6	32.77 min	1.09
ImageNet10	10,400	3	29.35 min	0.94
COCO	20,000	5	90.00 min	0.90

The first-run totals in Tables 7 and 8 include the applicable source-side preprocessing, subset selection, repair training, and evaluation. Every **AttriSelect** entry conservatively includes a fresh assignment of the full source pool. Thus, these tables are upper bounds for repeated repair cycles on the same task-level vocabulary.

Table 7: KITTI first-repair wall-clock time. Every **AttriSelect** row includes 32.77 minutes for fresh assignment of all 5,000 source images.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	**1.15 h**

Lightweight selectors remain cheaper at small ratios. At ratio 0.9, however, **AttriSelect** is both faster and more accurate than CoreSet and BADGE on COCO and KITTI, even after full-pool assignment is charged. For example, the COCO times are 1.72 hours for **AttriSelect**, 1.87 hours for CoreSet, and 3.01 hours for BADGE; their test AP values are 0.6355, 0.6202, and 0.6213, respectively. On KITTI, the corresponding times are 1.15, 1.63, and 1.42 hours, with mAP50–95 values 0.5850, 0.5781, and 0.5810.

Table 8: COCO first-repair wall-clock time. Every **AttriSelect** row includes 90 minutes for fresh assignment of all 20,000 source images.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	**1.72 h**

For a fixed task-level vocabulary, source assignment is a one-time preprocessing step. The same metadata can be reused across source ratios, attribute-aware selectors, and later repair cycles. After reuse, the **AttriSelect** totals equal the baseline training/evaluation totals plus less than two seconds of online selection: 42.20–48.89 minutes on KITTI and 4.44–19.68 minutes on COCO over ratios 0.1–0.9. Public metadata can remove this cost entirely when it covers the required vocabulary; CelebA, for example, provides 40 binary attributes with the dataset. Therefore, the 39–283× values in Table 2 describe source-subset selection speed after attribute records are available, not end-to-end pipeline speed.

C AttriSelect Algorithm

Algorithm 1: AttriSelect Source Selection for Targeted Repair

Require: Source pool $\mathcal{D}_{\text{src}} = \{(x_i, y_i, A_i)\}_{i=1}^{N}$ with VLM-assigned attribute records $A_i = \{v_{i,a} : a \in \mathcal{A}\}$; generated repair pool $\mathcal{D}_{\text{gen}}$ with attribute records; source-selection budget n_{select}; score parameters α, λ, w; numerical constant ε

Ensure: Selected source subset $\mathcal{D}_{\text{sel}}^{\text{src}}$

Compute class-conditional source counts $n_a(v \mid c)$ for every attribute a, value v, and class c

Compute effective attribute counts $\hat{n}_a(v) = n_a^{\text{src}}(v) + \tilde{n}_a^{\text{gen}}(v)$ using the source pool and generated repair pool

for all $(x_i, y_i, A_i) \in \mathcal{D}_{\text{src}}$ **do**

  $R_{i,a} \leftarrow (n_a(v_{i,a} \mid y_i) + \varepsilon)^{-\alpha}$ for each $a \in \mathcal{A}$

  $P_{i,a} \leftarrow -\lambda \cdot \hat{n}_a(v_{i,a}) / \sum_{v \in \mathcal{V}_a} \hat{n}_a(v)$ for each $a \in \mathcal{A}$

  $\bar{s}_i \leftarrow \sum_{a \in \mathcal{A}} (R_{i,a} + P_{i,a})$

end for

$\mathcal{D}_{\text{sel}}^{\text{src}} \leftarrow \emptyset$; initialize selected class counts $n_c \leftarrow 0$ for all classes

Set the soft per-class target $t \leftarrow n_{\text{select}} / N_{\text{cls}}$

while $|\mathcal{D}_{\text{sel}}^{\text{src}}| < n_{\text{select}}$ **do**

  for all remaining candidates $x_i \notin \mathcal{D}_{\text{sel}}^{\text{src}}$ with class $y_i = c$ **do**

    $C_c \leftarrow w \cdot (t - n_c)/t$

    $\tilde{s}_i \leftarrow \bar{s}_i + C_c$

  end for

  Select $x_j \leftarrow \arg\max_{x_i \notin \mathcal{D}_{\text{sel}}^{\text{src}}} \tilde{s}_i$

  $\mathcal{D}_{\text{sel}}^{\text{src}} \leftarrow \mathcal{D}_{\text{sel}}^{\text{src}} \cup \{x_j\}$; update $n_{y_j} \leftarrow n_{y_j} + 1$

end while

return $\mathcal{D}_{\text{sel}}^{\text{src}}$

The generated repair pool is fixed during selection. It affects **AttriSelect** only through the effective attribute counts in the frequency-penalty term, so all compared selectors use the same generated-data budget.

D Attribute Assignment Details

CelebA and ImageNet10. Attribute assignments for CelebA use the official CelebA annotations and follow the SafeFix setting [40]. ImageNet10 does not provide official attribute annotations, so we follow the SafeFix-style ImageNet10 protocol: define task-level attribute fields such as `color`, `background`, and `texture`, then assign closed-set attribute values to each image with the VLM. These assignments are used only as selection metadata for **AttriSelect** and for post-training slice diagnostics; benchmark labels remain the source of the reported task metrics.

COCO and KITTI. For COCO pose estimation and KITTI object detection, we follow HiBug2 [8] and use task-specific closed-set attribute fields proposed before source selection. We take KITTI as a concrete example. GPT-5 [29] proposes 10 representative object-detection categories, listed in Figure 3. Each category has a predefined set of candidate values, a natural-language prompt template, and a corresponding query directed to Qwen3-VL [3]. The repair fields used in the main KITTI experiments are color and weather; the remaining fields are retained for broader slice analysis and diagnostics.

Figure 3: KITTI object-detection attribute corpus used for VLM assignment.

KITTI Attribute Corpus & VLM Prompts

Note: The placeholder `#1` is replaced by a candidate value from the corresponding category.

`color`: white, black, gray, silver, red, blue, green, yellow, brown
Prompt: `a #1 main target object in a driving scene`

`object_scale`: small, medium, large
Prompt: `a #1 target object in a driving scene`

`viewpoint`: front, rear, side, front-side, rear-side
Prompt: `a #1 view of the main target object in a driving scene`

`occlusion`: none, partial, heavy
Prompt: `a #1 occluded target object in a driving scene`

`truncation`: none, partial, heavy
Prompt: `a #1 truncated target object in a driving scene`

`motion_state`: moving, stationary
Prompt: `a #1 target object in a driving scene`

`background_style`: city street, residential street, highway, parking area, service lane, tree-lined road
Prompt: `a driving scene on a #1`

`traffic_density`: sparse, light, moderate, heavy
Prompt: `a driving scene with #1 traffic`

`lighting_condition`: bright, overcast, strong shadows, glare, dim
Prompt: `a driving scene with #1 lighting`

`weather`: clear, cloudy, rainy, foggy, snowy
Prompt: `a driving scene in #1 weather`

E Main-Result Metric Tables

This section reports the original task metrics corresponding to the RER tables in the main text. The metrics are test accuracy for ImageNet10 and CelebA, COCO keypoint AP for pose estimation, and test mAP50–95 for KITTI object detection. All entries report the mean and standard deviation ($\pm$) over seeds 0, 42, 1234, 3407, and 2024. We performed paired t-tests to evaluate statistical significance, confirming that the improvements of **AttriSelect** over the respective best baselines are statistically significant ($p < 0.05$) across the reported selection ratios.

Table 9: ImageNet10 classification repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	$0.7269_{\pm 0.0025}$	$0.7271_{\pm 0.0037}$	$0.7254_{\pm 0.0016}$	$0.7269_{\pm 0.0032}$	$0.7323_{\pm 0.0022}$	$0.7331_{\pm 0.0023}$	$0.7285_{\pm 0.0026}$	$\mathbf{0.7369}_{\pm 0.0043}$
0.3	$0.7308_{\pm 0.0020}$	$0.7369_{\pm 0.0036}$	$0.7292_{\pm 0.0016}$	$0.7369_{\pm 0.0012}$	$0.7423_{\pm 0.0036}$	$0.7469_{\pm 0.0043}$	$0.7392_{\pm 0.0022}$	$\mathbf{0.7492}_{\pm 0.0044}$
0.5	$0.7380_{\pm 0.0030}$	$0.7446_{\pm 0.0017}$	$0.7331_{\pm 0.0027}$	$0.7400_{\pm 0.0017}$	$0.7469_{\pm 0.0029}$	$0.7410_{\pm 0.0028}$	$0.7423_{\pm 0.0016}$	$\mathbf{0.7508}_{\pm 0.0010}$
0.7	$0.7423_{\pm 0.0043}$	$0.7346_{\pm 0.0043}$	$0.7354_{\pm 0.0042}$	$0.7392_{\pm 0.0043}$	$0.7477_{\pm 0.0011}$	$0.7469_{\pm 0.0034}$	$0.7423_{\pm 0.0034}$	$\mathbf{0.7515}_{\pm 0.0037}$
0.9	$0.7462_{\pm 0.0040}$	$0.7469_{\pm 0.0011}$	$0.7462_{\pm 0.0026}$	$0.7346_{\pm 0.0035}$	$0.7438_{\pm 0.0027}$	$0.7500_{\pm 0.0033}$	$0.7508_{\pm 0.0041}$	$\mathbf{0.7562}_{\pm 0.0019}$
1.0	$0.7400_{\pm 0.0013}$	$0.7400_{\pm 0.0042}$	$0.7400_{\pm 0.0020}$	$0.7400_{\pm 0.0022}$	$0.7400_{\pm 0.0039}$	$0.7400_{\pm 0.0011}$	$0.7400_{\pm 0.0040}$	$0.7400_{\pm 0.0027}$

Table 10: CelebA classification repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	$0.9103_{\pm 0.0034}$	$0.9080_{\pm 0.0040}$	$0.9124_{\pm 0.0020}$	$0.9000_{\pm 0.0042}$	$0.9087_{\pm 0.0022}$	$0.9047_{\pm 0.0027}$	$0.9113_{\pm 0.0035}$	$\mathbf{0.9137}_{\pm 0.0024}$
0.3	$0.9114_{\pm 0.0011}$	$0.9107_{\pm 0.0011}$	$0.9139_{\pm 0.0027}$	$0.9132_{\pm 0.0036}$	$0.9075_{\pm 0.0036}$	$0.9090_{\pm 0.0037}$	$0.9095_{\pm 0.0032}$	$\mathbf{0.9141}_{\pm 0.0014}$
0.5	$0.9114_{\pm 0.0010}$	$0.9132_{\pm 0.0016}$	$0.9129_{\pm 0.0021}$	$0.9134_{\pm 0.0042}$	$0.9084_{\pm 0.0044}$	$0.9078_{\pm 0.0013}$	$0.9151_{\pm 0.0011}$	$\mathbf{0.9187}_{\pm 0.0014}$
0.7	$0.9084_{\pm 0.0024}$	$0.9135_{\pm 0.0027}$	$0.9153_{\pm 0.0016}$	$0.9119_{\pm 0.0035}$	$0.9133_{\pm 0.0015}$	$0.9105_{\pm 0.0018}$	$0.9114_{\pm 0.0041}$	$\mathbf{0.9171}_{\pm 0.0012}$
0.9	$0.9087_{\pm 0.0041}$	$0.9100_{\pm 0.0014}$	$0.9135_{\pm 0.0039}$	$0.9114_{\pm 0.0012}$	$0.9114_{\pm 0.0015}$	$0.9100_{\pm 0.0038}$	$0.9119_{\pm 0.0013}$	$\mathbf{0.9196}_{\pm 0.0022}$
1.0	$0.9114_{\pm 0.0041}$	$0.9114_{\pm 0.0042}$	$0.9114_{\pm 0.0017}$	$0.9114_{\pm 0.0026}$	$0.9114_{\pm 0.0023}$	$0.9114_{\pm 0.0038}$	$0.9114_{\pm 0.0042}$	$0.9114_{\pm 0.0028}$

F Additional Controlled Repair Comparisons

The main tables compare **AttriSelect** with generic source-selection methods. This section adds controls that receive the same repair-relevant attributes and target definition, a generated-distribution complement control, exact-conjunction evaluation, selected-subset diagnostics, and more constrained source budgets. Unless stated otherwise, task metrics are five-seed means over seeds 0, 42, 1234, 3407, and 2024. Ratio Mean is the arithmetic mean of the displayed ratio-level means.

F.1 Same-Information Baselines

We evaluate three direct attribute-aware controls. **Target-Slice + Stratified** selects exact target-conjunction examples first and fills the remaining budget with class-stratified background samples. **Attribute-Filtered Random** prioritizes examples matching at least one target field, samples from class-balanced queues, and then fills from the background. **Attribute-Filtered Entropy** uses the same candidate pool and queues but ranks candidates by diagnosis-checkpoint entropy. All methods use the same fixed 1,000-image generated set, source budget, checkpoint, repair schedule, and attribute metadata.

AttriSelect has the highest Ratio Mean on all three tasks and the highest RER at 14 of the 15 task–ratio points. Generated-Complement is higher only on KITTI at ratio 0.7, by 0.54 RER points. The comparison separates access to repair-relevant attributes from the full score: direct attribute filtering helps, but it does not reproduce the aggregate ordering of **AttriSelect**.

F.2 Generated-Complement Control

Generated-Complement tests whether generated-set complementarity and hard group balance alone explain the result. For source sample x, let $f_{\mathrm{gen},a}(v)$ be the number of generated examples with value v for target field a. The control uses

$$S_{\mathrm{GC}}(x) = \frac{1}{|\mathcal{A}_x|} \sum_{a \in \mathcal{A}_x} \frac{1}{1 + f_{\mathrm{gen},a}(x_a)}, \tag{8}$$

followed by deterministic round-robin balancing over task-specific groups. It does not use source rarity, diagnostic weights, uncertainty, or the full $R + P + C$ score.

On matched selected subsets, Generated-Complement has class imbalance no worse than **AttriSelect** at all 15 task–ratio points and lower imbalance at 11. Its mean complement score is higher at 14 points and tied at COCO ratio 0.1. The proxy objectives therefore behave as intended, but Table 15 shows lower aggregate repair. Generated-set complementarity and hard group balance alone do not explain the **AttriSelect** result.

Table 11: COCO pose-estimation repair results. Metric: COCO keypoint AP.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	$0.6048_{\pm 0.0028}$	$0.6110_{\pm 0.0032}$	$0.6132_{\pm 0.0019}$	$0.5877_{\pm 0.0040}$	$0.5985_{\pm 0.0039}$	$0.6013_{\pm 0.0044}$	$0.5950_{\pm 0.0015}$	$\mathbf{0.6183}_{\pm 0.0044}$
0.3	$0.6040_{\pm 0.0030}$	$0.6243_{\pm 0.0021}$	$0.6215_{\pm 0.0029}$	$0.6256_{\pm 0.0031}$	$0.6153_{\pm 0.0044}$	$0.6079_{\pm 0.0025}$	$0.6165_{\pm 0.0020}$	$\mathbf{0.6306}_{\pm 0.0022}$
0.5	$0.6209_{\pm 0.0020}$	$0.6230_{\pm 0.0027}$	$0.6282_{\pm 0.0026}$	$0.6167_{\pm 0.0036}$	$0.6206_{\pm 0.0018}$	$0.6153_{\pm 0.0038}$	$0.6182_{\pm 0.0022}$	$\mathbf{0.6317}_{\pm 0.0012}$
0.7	$0.6151_{\pm 0.0041}$	$0.6299_{\pm 0.0021}$	$0.6293_{\pm 0.0022}$	$0.6275_{\pm 0.0022}$	$0.6256_{\pm 0.0016}$	$0.6155_{\pm 0.0041}$	$0.6223_{\pm 0.0044}$	$\mathbf{0.6319}_{\pm 0.0024}$
0.9	$0.6091_{\pm 0.0040}$	$0.6291_{\pm 0.0021}$	$0.6170_{\pm 0.0018}$	$0.6202_{\pm 0.0023}$	$0.6213_{\pm 0.0028}$	$0.6191_{\pm 0.0014}$	$0.6301_{\pm 0.0025}$	$\mathbf{0.6355}_{\pm 0.0031}$
1.0	$0.6259_{\pm 0.0045}$	$0.6259_{\pm 0.0037}$	$0.6259_{\pm 0.0012}$	$0.6259_{\pm 0.0040}$	$0.6259_{\pm 0.0017}$	$0.6259_{\pm 0.0015}$	$0.6259_{\pm 0.0011}$	$0.6259_{\pm 0.0033}$

Table 12: KITTI object-detection repair results. Metric: test mAP50–95.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	$0.5204_{\pm 0.0021}$	$0.5308_{\pm 0.0037}$	$0.5297_{\pm 0.0037}$	$0.5237_{\pm 0.0039}$	$0.5322_{\pm 0.0025}$	$0.5222_{\pm 0.0031}$	$0.5262_{\pm 0.0036}$	$\mathbf{0.5372}_{\pm 0.0023}$
0.3	$0.5605_{\pm 0.0017}$	$0.5572_{\pm 0.0036}$	$0.5636_{\pm 0.0033}$	$0.5612_{\pm 0.0013}$	$0.5624_{\pm 0.0030}$	$0.5573_{\pm 0.0021}$	$0.5568_{\pm 0.0039}$	$\mathbf{0.5646}_{\pm 0.0023}$
0.5	$0.5719_{\pm 0.0030}$	$0.5680_{\pm 0.0039}$	$0.5766_{\pm 0.0044}$	$0.5759_{\pm 0.0043}$	$0.5726_{\pm 0.0013}$	$0.5675_{\pm 0.0027}$	$0.5721_{\pm 0.0021}$	$\mathbf{0.5773}_{\pm 0.0012}$
0.7	$0.5766_{\pm 0.0039}$	$0.5760_{\pm 0.0013}$	$0.5772_{\pm 0.0024}$	$0.5772_{\pm 0.0013}$	$0.5717_{\pm 0.0017}$	$0.5768_{\pm 0.0024}$	$\mathbf{0.5782}_{\pm 0.0030}$	$0.5780_{\pm 0.0039}$
0.9	$0.5817_{\pm 0.0039}$	$0.5830_{\pm 0.0037}$	$0.5830_{\pm 0.0042}$	$0.5781_{\pm 0.0011}$	$0.5810_{\pm 0.0018}$	$0.5838_{\pm 0.0044}$	$0.5796_{\pm 0.0026}$	$\mathbf{0.5850}_{\pm 0.0012}$
1.0	$0.5858_{\pm 0.0021}$	$0.5858_{\pm 0.0024}$	$0.5858_{\pm 0.0031}$	$0.5858_{\pm 0.0026}$	$0.5858_{\pm 0.0020}$	$0.5858_{\pm 0.0043}$	$0.5858_{\pm 0.0026}$	$0.5858_{\pm 0.0014}$

F.3 Exact Diagnosed Conjunctions

The attribute heatmaps in the main paper report marginal values. We separately evaluate the full diagnosed conjunction and its strict complement at source ratio 0.5. Table 16 reports the strongest controls and **AttriSelect**; all values are mean $\pm$ sample standard deviation over five seeds.

Relative to the common pre-repair checkpoint, **AttriSelect** improves the ImageNet10 exact conjunction by 13.75 percentage points and its non-target complement by 3.22 points. The COCO gains are 10.13 and 4.04 points. For the 16-image ImageNet10 slice, Generated-Complement records per-seed correct counts $12/11/12/13/12$ and **AttriSelect** records $13/12/13/11/12$, for totals of $60/80$ and $61/80$. We therefore treat the exact-target method ordering as descriptive. The supported conclusion is that the overall gain includes the diagnosed conjunction while performance on the remaining test images also increases.

F.4 Selected-Subset Composition

We analyze ImageNet10 selected subsets using unique $(\text{class}, \text{color}, \text{background}, \text{texture})$ tuples and mean categorical Hamming distance over color, background, and texture. Table 17 compares **AttriSelect** with the best direct control at each ratio; the best control is selected independently for each metric from Target-Slice + Stratified, Attribute-Filtered Random, Attribute-Filtered Entropy, and Generated-Complement.

AttriSelect covers more class–attribute tuples and has larger mean pairwise attribute distance at every ratio. Generated-Complement has lower redundancy with the generated target values but lower RER, so avoiding generated-target duplication alone is insufficient. The combined evidence is consistent with retaining repair-relevant samples while preserving broader class and attribute coverage.

F.5 More Constrained Source Budgets

We further reduce the source ratios to 0.02 and 0.05, corresponding to 208/520 selected ImageNet10 images and 400/1,000 selected COCO images. The diagnosis checkpoint, generated repair set, repair schedule, and five seeds remain fixed.

Many baselines have negative RER in this regime, showing that the fixed source/generated mixture and repair schedule can be inadequate at very small source budgets. Under the matched protocol, **AttriSelect** is highest at all four dataset–ratio points. Its two-ratio mean exceeds the strongest alternative by 1.50 RER points on ImageNet10 and 1.34 points on COCO.

G Attribute-Slice Heatmaps

Figure 4 reports the corresponding ratio-0.5 COCO pose-estimation slice analysis over the full annotated attribute vocabulary. The targeted repair values are clothes color brown and rainy weather.

Table 13: ImageNet10 same-information repair comparison. Metric: RER (%; five-seed mean ± standard deviation).

Ratio	CoreSet	BADGE	UHerding	PPAL	Target+Strat.	Filtered Rand.	Filtered Ent.	Gen.-Comp.	Ours
0.1	$3.40_{\pm 1.13}$	$5.31_{\pm 0.78}$	$5.59_{\pm 0.81}$	$3.96_{\pm 0.92}$	$4.60_{\pm 1.80}$	$5.91_{\pm 1.20}$	$5.09_{\pm 0.88}$	$1.50_{\pm 0.87}$	$\mathbf{6.93}_{\pm 1.52}$
0.3	$6.93_{\pm 0.42}$	$8.84_{\pm 1.27}$	$10.47_{\pm 1.52}$	$7.75_{\pm 0.78}$	$7.50_{\pm 0.85}$	$7.39_{\pm 1.49}$	$6.79_{\pm 0.81}$	$5.31_{\pm 1.04}$	$\mathbf{11.28}_{\pm 1.56}$
0.5	$8.03_{\pm 0.60}$	$10.47_{\pm 1.03}$	$8.38_{\pm 0.99}$	$8.84_{\pm 0.57}$	$9.37_{\pm 1.31}$	$9.52_{\pm 1.34}$	$9.52_{\pm 1.10}$	$11.29_{\pm 0.76}$	$\mathbf{11.85}_{\pm 0.35}$
0.7	$7.75_{\pm 1.52}$	$10.75_{\pm 0.39}$	$10.47_{\pm 1.20}$	$8.84_{\pm 1.20}$	$9.23_{\pm 2.30}$	$10.65_{\pm 1.34}$	$10.93_{\pm 0.71}$	$8.57_{\pm 1.18}$	$\mathbf{12.10}_{\pm 1.31}$
0.9	$6.12_{\pm 1.24}$	$9.37_{\pm 0.96}$	$11.57_{\pm 1.17}$	$11.85_{\pm 1.45}$	$11.04_{\pm 0.92}$	$10.47_{\pm 1.27}$	$11.67_{\pm 1.59}$	$10.48_{\pm 0.95}$	$\mathbf{13.76}_{\pm 0.67}$
Mean	6.44	8.95	9.30	8.25	8.35	8.79	8.80	7.43	**11.19**

Table 14: COCO and KITTI same-information repair comparison. Metric: RER (%; five-seed mean). Each cell reports COCO / KITTI.

Ratio	CoreSet	BADGE	UHerding	PPAL	Target+Strat.	Filtered Rand.	Filtered Ent.	Gen.-Comp.	Ours
0.1	-0.61/-8.89	2.03/-6.95	2.71/-9.24	1.17/-8.32	5.63/-8.24	-3.42/-6.14	4.47/-6.81	2.50/-6.24	**6.86/-5.81**
0.3	8.64/-0.32	6.12/-0.05	4.32/-1.21	6.42/-1.33	9.38/-0.62	6.53/0.04	8.37/0.13	7.22/0.17	**9.86/0.46**
0.5	6.47/3.04	7.42/2.29	6.12/1.12	6.83/2.17	8.83/2.83	7.11/2.58	8.16/3.25	8.67/1.72	**10.13/3.36**
0.7	9.10/3.34	8.64/2.08	6.17/3.25	7.83/3.57	9.24/3.54	9.26/2.93	9.68/3.43	9.56/**4.06**	**10.18**/3.52
0.9	7.32/3.54	7.59/4.21	7.05/4.85	9.74/3.89	8.68/4.38	6.31/4.46	9.78/4.68	6.96/3.74	**11.05/5.12**
Mean	6.18/0.14	6.36/0.32	5.27/-0.25	6.40/0.00	8.35/0.38	5.16/0.77	8.09/0.94	6.98/0.69	**9.62/1.33**

As in the ImageNet10 example, the largest gains appear on the targeted values, while the remaining attributes show smaller changes.

H Effect of the VLM and Human Audit

VLM Role and Leakage Control **Where VLM annotations enter.** **AttriSelect** uses the VLM as a fixed attribute annotator. Given a task-specific set of attribute fields and closed-set label options, the VLM assigns a discrete value to each image. The selector then uses these values, class labels, and generated-data attribute counts to choose the source subset. The repair-slice indicator is the binary conjunction of the target attribute values for the corresponding task: red hair, brown skin, and sad emotion for CelebA; pink color and fabric texture for ImageNet10; brown clothes and rainy weather for COCO; and red object color and cloudy weather for KITTI. The main repair metrics in Tables 1 and 9–12 are standard task metrics computed from benchmark labels, so the VLM does not grade the final repaired model. Test-set attributes are used only for post-training slice diagnostics.

Leakage control. The attribute vocabulary, target fields, and VLM annotation prompt are fixed before running the source-selection methods. The assignment stage does not receive model predictions, failure rates, method names, selected-set membership, or downstream results. In settings where the same VLM family is used for target-slice wording and image-level assignment, the two roles are still separated: the slice description is fixed first, and assignment is then run as a frozen closed-set classifier over images. Source selection uses only source-training attributes and generated-pool attributes. Thus a possible failure mode is not test-label leakage, but inaccurate or self-consistent metadata. The audit below tests the target fields with independent human labels.

Human Audit Protocol We conduct a blinded human audit with five AI graduate students. Each annotator labels 100 CelebA images, 100 COCO images, and 100 KITTI images using the same closed-set options as the VLM. Annotators see only the image, dataset name, and sample identifier; they do not see VLM labels, target flags, model predictions, data split, or method outputs. We aggregate human labels by majority vote and compare them with the VLM labels at two levels: exact agreement on the audited fields and binary agreement on the repair-slice indicator after applying the target values for those fields. For example, the COCO slice indicator is true only when the audited image has brown clothes and rainy weather, and the KITTI indicator is true only when it has red object color and cloudy weather. For CelebA, the indicator uses the full repair conjunction over red hair, brown skin, and sad emotion, rather than a single field such as hair color alone. The binary agreement is the most relevant check for **AttriSelect** because the selector uses these attributes to identify repair-relevant source examples.

The audit is designed to measure agreement rather than to estimate the prevalence of rare target conjunctions. As shown in Table 19, the VLM and human consensus usually agree on the audited

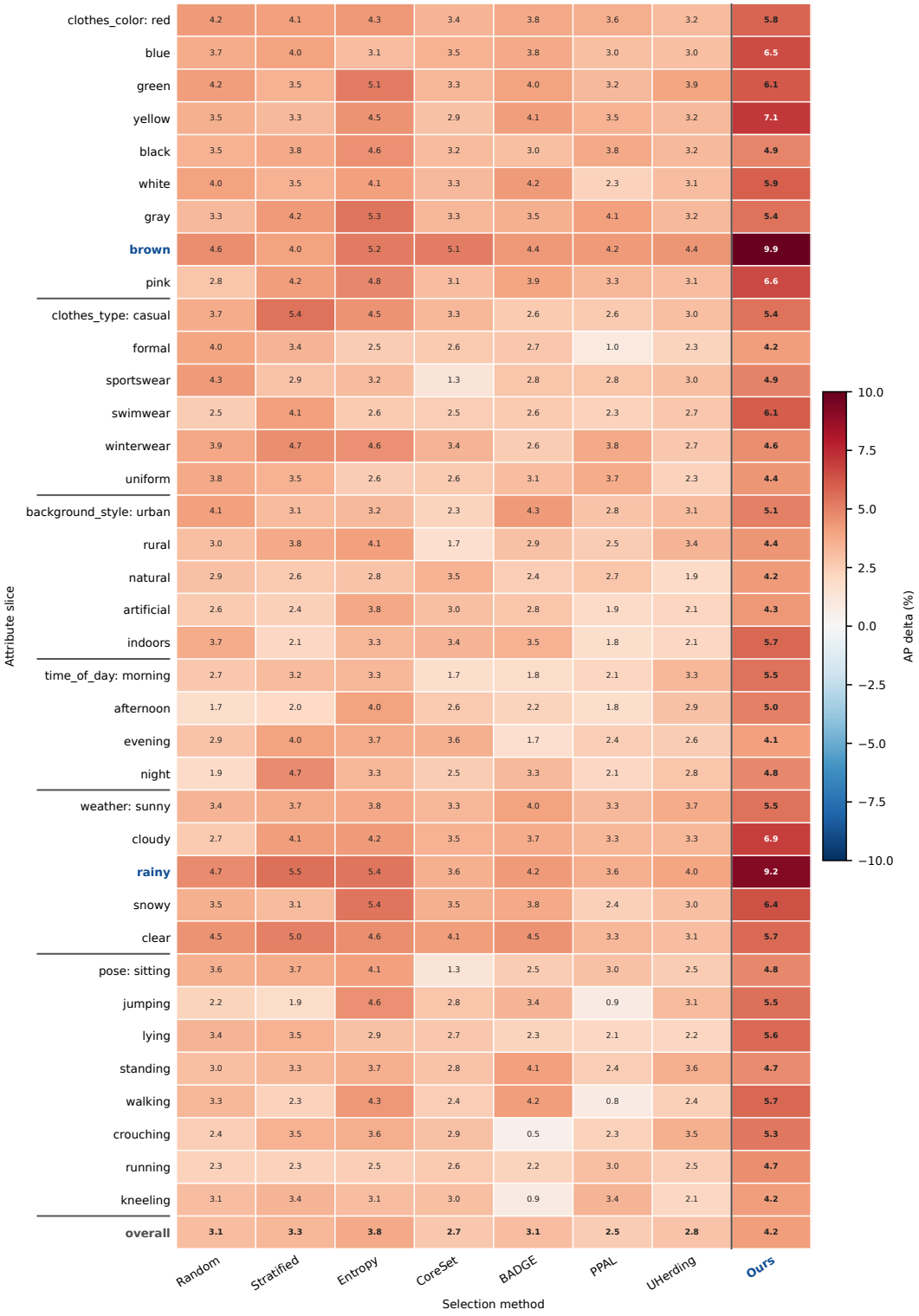


Figure 4: COCO pose-estimation ratio-0.5 attribute-slice AP deltas. The figure expands the slice-level analysis to all annotated COCO pose-estimation attribute groups. The targeted repair values are clothes color brown and rainy weather.

Table 15: Generated-Complement summary. Exact-target, non-target, and overall entries use each task's native metric. RER is the five-ratio mean. COCO and KITTI exact conjunctions contain only two test images per seed and are descriptive.

Task	Exact target	Non-target	Overall	RER mean	Ours RER mean
ImageNet10	0.7155	0.7392	0.7383	7.43	**11.19**
COCO	0.7261	0.6190	0.6188	6.98	**9.62**
KITTI	0.4257	0.5667	0.5664	0.69	**1.33**

Table 16: Exact-conjunction and non-target performance at source ratio 0.5. ImageNet10 uses accuracy on "pink AND fabric" ($n = 16$) and its complement ($n = 1{,}284$). COCO uses keypoint AP on "brown AND rainy" ($n = 22$) and its complement ($n = 1{,}151$).

Method	IN exact	IN non-target	IN overall	COCO exact	COCO non-target	COCO overall
Pre-repair	0.6250	0.7185	0.7173	0.4813	0.5923	0.5902
Target+Stratified	$0.7375_{\pm 0.0523}$	$0.7473_{\pm 0.0021}$	$0.7472_{\pm 0.0020}$	$0.5696_{\pm 0.0074}$	$0.6279_{\pm 0.0026}$	$0.6268_{\pm 0.0025}$
Filtered Random	$0.7375_{\pm 0.0685}$	$0.7477_{\pm 0.0025}$	$0.7476_{\pm 0.0024}$	$0.5524_{\pm 0.0076}$	$0.6229_{\pm 0.0032}$	$0.6216_{\pm 0.0031}$
Filtered Entropy	$0.7375_{\pm 0.0815}$	$0.7483_{\pm 0.0023}$	$0.7482_{\pm 0.0022}$	$0.5629_{\pm 0.0035}$	$0.6269_{\pm 0.0017}$	$0.6257_{\pm 0.0016}$
Generated-Complement	$0.7500_{\pm 0.0442}$	$0.7461_{\pm 0.0019}$	$0.7461_{\pm 0.0018}$	$0.5680_{\pm 0.0079}$	$0.6265_{\pm 0.0030}$	$0.6254_{\pm 0.0029}$
Ours	$\mathbf{0.7625}_{\pm 0.0523}$	$\mathbf{0.7507}_{\pm 0.0011}$	$\mathbf{0.7508}_{\pm 0.0010}$	$\mathbf{0.5826}_{\pm 0.0029}$	$\mathbf{0.6326}_{\pm 0.0013}$	$\mathbf{0.6317}_{\pm 0.0012}$

target fields and on the resulting binary slice indicators. This supports the use of the VLM as a scalable attribute annotator rather than as an oracle: the VLM provides metadata, and the human audit checks that the target fields used for repair are visually grounded.

Sensitivity to attribute noise. We also test whether **AttriSelect** is brittle to noisy attributes. For each audited dataset and each target field, we independently corrupt a fraction p of VLM labels by replacing the original value with a different valid value from the same field vocabulary. We then rerun source selection with the corrupted attributes while keeping the generated repair set, source ratio, and training recipe fixed. Table 20 reports the change from the clean-VLM **AttriSelect** result at source ratio 0.5 for CelebA, COCO, and KITTI. Small changes under moderate noise would indicate that the selector depends on the aggregate attribute distribution rather than on a small number of individual VLM decisions.

The results show that **AttriSelect** is relatively stable to light and moderate attribute noise. At 5% and 10% corruption, the metric changes are small across all three datasets, which suggests that the selector is not driven by a few isolated attribute assignments. The larger shifts at 20% noise, especially on CelebA and KITTI, are expected: once many target-field labels are replaced, the rarity and frequency counts no longer describe the diagnosed repair slice reliably. In this high-noise regime, **AttriSelect** no longer behaves like a clean semantic selector. Its ranking is partly driven by corrupted attribute values, while the soft class-balance term still preserves label coverage, so the selected subset moves toward a mixture of noisy attribute sampling and soft stratified sampling. This explains why the results begin to approach the behavior of random or stratified source selection when the attribute metadata is heavily corrupted. COCO is less sensitive in this range, suggesting that its selected subset is more strongly supported by aggregate attribute statistics or by other correlated visual factors in the source pool.

I Effect of the Repair Pipeline and Image Generation

I.1 Alternative Repair Pipeline

The main experiments use a SafeFix-style upstream workflow. To test whether the source-selection result depends on that pipeline, we replace its generated repair set with a fixed HiBug-generated set on CelebA. All selectors share the diagnosis checkpoint, 80,000-image source pool, generated-image budget, source ratios, retraining configuration, and five seeds. Only the source-selection rule changes.

AttriSelect has the highest RER at all five ratios. Its margins over the strongest baseline are $0.44/0.05/2.90/1.13/5.47$ RER points, or 2.00 points on average. This comparison is internal to the HiBug table: changing the upstream repair pipeline changes the generated set and can shift every selector's absolute RER.

Table 17: ImageNet10 selected-subset mechanism summary. RER values are five-seed means; subset diagnostics use matched selections. Higher is better except for class imbalance.

	RER		Unique tuples		Attribute distance		Class imbalance	
Ratio	Best ctrl.	Ours	Best ctrl.	Ours	Best ctrl.	Ours	Best ctrl.	Ours
0.02	0.68	**1.23**	142	**184**	0.7695	**0.8823**	**1**	**1**
0.05	3.95	**5.58**	267	**397**	0.7828	**0.8680**	**0**	**0**
0.10	5.91	**6.93**	410	**680**	0.7870	**0.8597**	**0**	**0**
0.30	7.50	**11.28**	801	**1,165**	0.7902	**0.8414**	**0**	2
0.50	11.29	**11.85**	1,007	**1,318**	0.7912	**0.8272**	**0**	4
0.70	10.93	**12.10**	1,194	**1,379**	0.7919	**0.8124**	**1**	5
0.90	11.67	**13.76**	1,345	**1,403**	0.7915	**0.7977**	**0**	6

Table 18: Repair under constrained source budgets. Metric: RER (%; five-seed mean ± standard deviation).

Task	Ratio	Random	Strat.	Entropy	CoreSet	BADGE	UHerd.	PPAL	Gen.-Comp.	Target+Strat.	Filt. Rand./Ent.	Ours
ImageNet10	0.02	-3.10$_{\pm 1.22}$	-2.68$_{\pm 1.08}$	-2.35$_{\pm 1.34}$	-3.42$_{\pm 1.16}$	-2.12$_{\pm 1.27}$	-2.57$_{\pm 1.05}$	-2.89$_{\pm 1.18}$	-1.77$_{\pm 1.21}$	-2.04$_{\pm 1.38}$	0.68$_{\pm 0.94}$/-0.95$_{\pm 1.07}$	**1.23**$_{\pm 0.72}$
ImageNet10	0.05	1.84$_{\pm 0.91}$	2.41$_{\pm 0.84}$	2.76$_{\pm 1.02}$	1.58$_{\pm 0.88}$	2.91$_{\pm 0.95}$	2.64$_{\pm 0.90}$	2.20$_{\pm 0.82}$	3.13$_{\pm 0.86}$	3.95$_{\pm 0.92}$	3.13$_{\pm 0.78}$/2.32$_{\pm 0.81}$	**5.58**$_{\pm 0.61}$
COCO	0.02	-8.75$_{\pm 1.42}$	-7.15$_{\pm 1.31}$	-5.62$_{\pm 1.18}$	-9.30$_{\pm 1.57}$	-6.40$_{\pm 1.26}$	-8.10$_{\pm 1.49}$	-7.00$_{\pm 1.35}$	-2.84$_{\pm 1.16}$	-7.80$_{\pm 1.43}$	-2.46$_{\pm 1.02}$/-6.23$_{\pm 1.28}$	**-2.02**$_{\pm 0.91}$
COCO	0.05	-4.20$_{\pm 1.12}$	-2.96$_{\pm 0.98}$	-1.84$_{\pm 0.87}$	-5.10$_{\pm 1.24}$	-2.55$_{\pm 0.93}$	-4.00$_{\pm 1.15}$	-3.20$_{\pm 1.01}$	-31.00$_{\pm 2.34}$	-11.64$_{\pm 1.76}$	-0.61$_{\pm 0.74}$/-1.10$_{\pm 0.82}$	**1.63**$_{\pm 0.68}$

I.2 Alternative Image Generators

The upstream workflow following SafeFix [40] uses ControlNet [43] as the conditional diffusion model. The quality of diffusion-based dataset augmentation can vary with the generator and editing procedure [35, 10, 37]. To test whether **AttriSelect** depends on this particular generator, we replace ControlNet while keeping the rest of the repair pipeline fixed: InstructPix2Pix [5] for CelebA and ICEdit [44] for COCO.

Table 22 reports CelebA results with InstructPix2Pix generation. **AttriSelect** remains the strongest source selector at every active source ratio. Its absolute RER is lower than the corresponding ControlNet result in Table 1, but the same decrease is visible across the other selectors, indicating that this shift comes from the generated repair images rather than from the source-selection rule.

Table 23 reports the corresponding COCO result with ICEdit. ICEdit improves the absolute RER values for most source ratios, and **AttriSelect** remains the best selector under this generator as well. These comparisons should be read horizontally: for a fixed generator and generated repair set, **AttriSelect** is compared against the non-attribute active-learning selectors. Changes in generator quality can shift the absolute repair numbers, but they do not remove the benefit of attribute-driven source selection.

J Additional Robustness and Sensitivity Studies

J.1 Repair-Schedule Sensitivity

The main ImageNet10 repair schedule uses 15 epochs, half of the 30-epoch diagnosis schedule. We vary repair training from 15 to 60 epochs at source ratios 0.5 and 0.9 while holding the diagnosis checkpoint, generated repair set, source budget, learning rate, and five seeds fixed.

AttriSelect is strongest in all eight ratio–epoch configurations. Its RER rises from 11.85 to 14.08 at ratio 0.5 and from 13.76 to 15.22 at ratio 0.9 as repair training increases from 15 to 60 epochs. The ordering is therefore not an artifact of the shorter submitted repair schedule.

J.2 Slice-Granularity Sensitivity

Slice granularity changes both the fields used for source selection and the generated repair set. We compare the diagnosed two-field conjunction with single-field variants and a narrower three-field conjunction over five seeds.

The diagnosed two-field slice is best in 8 of the 10 dataset–ratio settings. On average, it exceeds COCO Brown-only, Rain-only, and the three-field slice by 0.24, 0.80, and 0.66 AP points. On ImageNet10, it exceeds Pink-only, Fabric-only, and the three-field slice by 0.98, 0.91, and 0.50

Table 19: Human audit of VLM-assigned attributes. Five AI graduate students label each audited image using the same closed-set options as the VLM. Exact field agreement measures agreement on the audited attribute values, while slice-indicator agreement measures agreement on the binary target conjunction used for repair.

Dataset	Target field 1 agree.	Target field 2 agree.	Slice-indicator agree.
CelebA	Hair color: 98%	Skin tone: 99%	92%
COCO	Clothes color: 97%	Weather: 91%	90%
KITTI	Object color: 98%	Weather: 96%	94%

Table 20: Sensitivity to corrupted VLM attributes at source ratio 0.5. Entries report the change from the clean-VLM **AttriSelect** result for each task metric.

Dataset	Metric	5% noise	10% noise	20% noise
CelebA	Test ACC Δ	0.007	0.01	0.09
COCO	Test AP Δ	0.002	0.003	0.02
KITTI	Test mAP50–95 Δ	0.003	0.004	0.06

accuracy points. These results indicate a support–specificity trade-off: a single-field slice can mix different failure causes, while a narrow conjunction can remove useful support.

J.3 Incomplete Attribute-Vocabulary Coverage

We remove one target field while holding the generated repair pool, source budget, checkpoint, and training schedule fixed. Parenthesized values are the selected-set Jaccard overlap with the full two-field selector.

Missing-field variants are lower than Full in 17 of the 20 dataset–field–ratio comparisons. The selected subset changes most at low budgets: at ratio 0.1, Jaccard overlap falls as low as 0.3904 on ImageNet10 and 0.2034 on COCO. Mean ImageNet10 accuracy changes are $-0.94/-0.81$ points when color/texture is removed; mean COCO AP changes are $-0.0091/-0.0065$ when color/weather is removed. The effect is not monotonic at every ratio, but incomplete coverage usually weakens task performance.

K Source-Only Data Selection

We evaluate **AttriSelect** outside model repair on ImageNet10. These experiments use no generated images, diagnosed slices, or repair checkpoints. A fresh ResNet-18 is trained on each selected subset, and every result is a five-seed mean. In this setting, the generated-data frequency term is omitted and selection uses the available source attributes and soft class balancing, as formalized in Appendix L.

K.1 One-Shot Subset Selection

We add two direct semantic baselines. **Attribute Rarity** ranks images by class-conditional rarity of their assigned attribute values and applies soft class balancing. **Semantic CoreSet** one-hot encodes the assigned values and runs class-balanced greedy k-center selection in that semantic representation.

AttriSelect has the highest accuracy at all five budgets. Its margins over the strongest alternative are $0.54/0.15/1.22/0.39/0.03$ accuracy points at ratios $0.1/0.3/0.5/0.7/0.9$.

K.2 Iterative Active Learning

All methods start from the same labeled set at ratio 0.1 and add labeled data over five acquisition rounds. Object–Attribute Entropy Reduction adapts the object–attribute query criterion to the setting in which attributes are already known and only the class label is queried.

AttriSelect-AL is strongest from ratios 0.3 through 0.6 and improves Curve Mean by 1.40 points over BADGE. The source-only comparison shows that the selection rule is not tied to checkpoint fine-tuning or a generated repair set.

Table 21: CelebA repair results with the HiBug repair pipeline. Metric: five-seed mean RER (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.59	2.33	6.92	-6.20	2.87	-1.36	5.58	**7.36**
0.3	5.91	5.08	8.02	7.79	1.66	3.39	3.87	**8.07**
0.5	5.76	7.70	7.48	7.90	2.53	2.13	9.64	**12.54**
0.7	2.57	8.06	10.03	6.19	7.86	4.96	5.90	**11.16**
0.9	2.84	4.28	7.95	5.71	5.80	4.19	6.37	**13.42**

Table 22: CelebA repair results with InstructPix2Pix generation. Metric: RER (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (InstructPix2Pix)	Ours (ControlNet)
0.1	4.23	1.95	6.41	-6.65	2.34	-1.78	5.12	**7.65**	8.48
0.3	5.34	4.72	7.98	7.21	1.02	2.81	3.45	**8.12**	8.91
0.5	5.56	7.23	6.78	7.54	2.12	1.56	9.12	**11.17**	13.79
0.7	2.12	7.67	9.34	5.78	7.34	4.23	5.34	**12.13**	12.09
0.9	2.45	3.92	7.56	5.34	5.23	3.87	5.91	**13.87**	14.74

K.3 Alternative Attribute Sources

We repeat the source-only evaluation with Qwen3-VL, BLIP, and InternVL3 assignments under the same closed-set vocabulary, training protocol, and five seeds.

No attribute source is strongest in both settings. InternVL3 has the highest One-Shot Mean, 0.31 points above Qwen3-VL. Qwen3-VL is strongest in active learning, exceeding InternVL3 by 1.76 points in Curve Mean and 2.57 points at ratio 0.6. The method works with all three sources, but annotation quality and representation affect the selected subset and downstream result.

L Additional Theory Analysis

Slice enrichment We use *slice enrichment* to describe whether a selected source subset contains more examples from the diagnosed target slice than would be expected from its class-conditional base rate.

Proposition 1 (Attribute-agnostic AL requires target alignment). *Consider any AL selector that, within class c, chooses B_c source examples using only non-attribute information h_i, such as uncertainty, logits, embeddings, losses, detector difficulty, coverage, or image distances, and does not use the attribute values $\{v_{i,a} : a \in \mathcal{A}\}$. Let $\mathcal{D}_{\mathrm{AL}}$ denote its selected source subset. If the target-slice indicator $\mathbf{1}[x_i \in T_c]$ is conditionally independent of h_i given class c, with ties and internal randomness independent of the target indicator, then*

$$\mathbb{E}\big[|\mathcal{D}_{\mathrm{AL}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c}.$$

Thus an attribute-agnostic AL method can enrich the target slice only when its non-attribute signal is additionally aligned with that slice.

Proof of Proposition 1. Conditional independence makes the target indicators exchangeable with respect to any ranking, batch construction, or randomized choice computed from h_i inside class c. The B_c selected positions therefore contain, in expectation, the same target-slice fraction as the class pool itself, namely $|T_c|/N_c$. This argument does not depend on the particular AL heuristic. Uncertainty sampling, margin sampling, loss-based selection, diversity sampling, CoreSet, BADGE, PPAL-style uncertainty-then-diversity selection, and Uncertainty Herding-style uncertainty coverage all fall under the claim whenever their selection variables omit the current attribute metadata and are not otherwise aligned with the target slice.

As a concrete coverage-based instance, let z_i be the representation used by a diversity or coverage selector, such as cluster-based Diversity, CoreSet in feature space, or BADGE in

Table 23: COCO pose-estimation repair results with ICEdit generation. Metric: RER (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (ICEdit)	Ours (ControlNet)
0.1	3.78	5.23	5.87	-0.34	2.34	3.12	1.56	**7.12**	6.86
0.3	3.92	8.67	8.12	9.12	6.56	4.67	6.89	**10.34**	9.86
0.5	7.91	8.45	9.67	6.91	7.82	6.54	7.23	**10.67**	10.13
0.7	6.54	10.12	9.98	9.56	9.12	6.71	8.23	**10.87**	10.18
0.9	5.12	9.87	6.98	7.82	8.12	7.54	10.12	**11.56**	11.05

Table 24: ImageNet10 repair-schedule sensitivity. Metric: five-seed mean RER (%).

Ratio	Method	15 epochs	30 epochs	45 epochs	60 epochs
0.5	Strongest baseline	10.47	10.74	11.84	12.11
	Ours	**11.85**	**12.39**	**13.51**	**14.08**
0.9	Strongest baseline	11.85	12.33	13.44	13.76
	Ours	**13.76**	**14.25**	**14.99**	**15.22**

gradient-embedding space. CoreSet-style k-center selection minimizes

$$\max_i \min_{j \in S} d(z_i, z_j), \tag{9}$$

so achieving objective value at most ϵ is equivalent to constructing an ϵ-cover of the source representations. For a finite set A in this representation space, define the empirical covering number

$$\mathcal{N}_\epsilon^{\text{emp}}(A) = \min\big\{|C| : C \subseteq A,\ A \subseteq \cup_{z \in C} B(z, \epsilon)\big\}. \tag{10}$$

If the target slice in class c has empirical covering number

$$q_c = \mathcal{N}_\epsilon^{\text{emp}}\big(\{z_i : x_i \in T_c\}\big), \tag{11}$$

then an ϵ-cover objective can cover the entire target region using only q_c selected target representatives. Consequently, when the non-target region still contains uncovered ϵ-scale cells, a coverage-based selector has an optimal solution with at most q_c selected target examples, even if $|T_c| \gg q_c$.

For this coverage-based instance, take $h_i = z_i$. By definition of the empirical covering number, there exists a subset $C_T \subseteq \{z_i : x_i \in T_c\}$ with $|C_T| = q_c$ such that every target-slice representation is within distance ϵ of some point in C_T. Selecting the corresponding q_c target examples therefore makes the whole target slice ϵ-covered. Any additional target example only refines coverage inside an already covered target region. If non-target ϵ-scale cells remain uncovered, a coverage or diversity objective can spend the remaining representatives outside T_c without increasing its maximum covering radius. Thus feature-space coverage does not force more than q_c target selections.

The same conclusion follows from the usual volumetric intuition. If a class region has effective dimension d, volume V, and the target subregion has volume V_T, then away from boundary effects an ϵ-cover requires on the order of $V/(v_d\epsilon^d)$ cells for the full class region but only $V_T/(v_d\epsilon^d)$ cells for the target region, where v_d is the volume of the unit d-ball. When the bug slice is semantically important but geometrically compact, $V_T \ll V$, so the number of representatives allocated by a geometry-only objective can be small. The objective allocates capacity to representation volume, not to semantic repair importance. □

Rarity and generated-data penalty We now show why the attribute score assigns strict priority to a rare target slice under explicit data conditions. For any target example $x_i \in T_c$ and non-target example $x_j \notin T_c$ from the same class c, define the attributes on which they differ as

$$\mathcal{D}_{ij} = \{a \in \mathcal{A} : v_{i,a} \neq v_{j,a}\}. \tag{12}$$

Let

$$A_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\left(\frac{1}{n_a(v_{i,a} \mid c) + \varepsilon} \right)^\alpha - \left(\frac{1}{n_a(v_{j,a} \mid c) + \varepsilon} \right)^\alpha \right] \tag{13}$$

Table 25: Slice-granularity sensitivity on COCO. Metric: keypoint AP. The diagnosed slice is brown clothes and rainy weather.

Ratio	Brown only	Rain only	Brown+rain+night	Brown+rain
0.1	0.6161	0.6174	0.6097	**0.6183**
0.3	0.6238	0.6224	0.6179	**0.6306**
0.5	0.6284	0.6217	0.6246	**0.6317**
0.7	0.6327	**0.6333**	0.6291	0.6319
0.9	0.6348	0.6132	0.6336	**0.6355**

Table 26: Slice-granularity sensitivity on ImageNet10. Metric: test accuracy. The diagnosed slice is pink color and fabric texture.

Ratio	Pink only	Fabric only	Pink+fabric+rainy	Pink+fabric
0.1	0.7223	0.7346	**0.7398**	0.7369
0.3	0.7369	0.7431	0.7414	**0.7492**
0.5	0.7377	0.7438	0.7429	**0.7508**
0.7	0.7485	0.7423	0.7462	**0.7515**
0.9	0.7500	0.7354	0.7491	**0.7562**

be the rarity advantage of the target example, and let

$$F_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} - \frac{\hat{n}_a(v_{j,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \right] \tag{14}$$

be the frequency-penalty disadvantage introduced by source and generated counts. If generated data has flooded the target value, then F_{ij} can be positive; if not, the penalty does not hurt the target example relative to x_j.

Proposition 2 (Rarity-dominates-penalty margin)**.** *Assume $\alpha > 0$ and $A_{ij} > 0$ for every target/non-target pair (x_i, x_j) within class c. Define*

$$\lambda_c^\star = \min_{x_i \in T_c,\, x_j \notin T_c,\, F_{ij} > 0} \frac{A_{ij}}{F_{ij}}, \tag{15}$$

with $\lambda_c^\star = \infty$ if no pair has $F_{ij} > 0$. For any $0 \le \lambda < \lambda_c^\star$, every target example in T_c has larger base score $\bar{s}_i$ than every non-target example from the same class. In particular, when each target attribute value is rarer than the competing non-target value, i.e., $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ for all $a \in \mathcal{D}_{ij}$, the condition $A_{ij} > 0$ holds automatically.

Proof. For a target/non-target pair (x_i, x_j) in class c, attributes outside $\mathcal{D}_{ij}$ contribute the same score to both examples and cancel. The remaining base-score gap is

$$\bar{s}_i - \bar{s}_j = A_{ij} - \lambda F_{ij}. \tag{16}$$

If $F_{ij} \le 0$, then $\bar{s}_i - \bar{s}_j \ge A_{ij} > 0$ for any $\lambda \ge 0$, so the frequency penalty does not overturn the rarity advantage. If $F_{ij} > 0$, then $\bar{s}_i - \bar{s}_j > 0$ whenever $\lambda < A_{ij}/F_{ij}$. Taking the minimum over all target/non-target pairs with $F_{ij} > 0$ gives the interval $0 \le \lambda < \lambda_c^\star$, under which every pair has a strictly positive score gap. Finally, if $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ and $\alpha > 0$, then each rarity term in A_{ij} is positive, so their sum is positive. □

This proposition is the main data-dependent guarantee. It says that if the VLM annotations correctly identify a genuinely rare target slice, then there is a nonempty range of λ values for which rarity dominates the synthetic-data frequency penalty. Even when generated examples make the target value more frequent in the repair candidate pool, the target examples remain ranked above non-target examples as long as λ is below the explicit threshold $\lambda_c^\star$. Conversely, if target and non-target samples are indistinguishable under $\mathcal{A}$, then no attribute-only selector can separate them without additional information.

Table 27: Missing-field sensitivity on ImageNet10. Metric: overall test accuracy (%).

Ratio	Full	Missing color	Missing texture
0.1	73.69	72.69 (0.5604)	**73.85** (0.3904)
0.3	**74.92**	73.77 (0.6166)	73.54 (0.5842)
0.5	**75.08**	74.23 (0.6443)	74.23 (0.7330)
0.7	**75.15**	74.00 (0.6999)	74.23 (0.8826)
0.9	**75.62**	75.08 (0.8765)	74.54 (0.9668)

Table 28: Missing-field sensitivity on COCO. Metric: keypoint AP.

Ratio	Full	Missing color	Missing weather
0.1	**0.6183**	0.6055 (0.4782)	0.6105 (0.2034)
0.3	**0.6306**	0.6221 (0.5611)	0.6214 (0.4613)
0.5	**0.6317**	0.6178 (0.7063)	0.6271 (0.6084)
0.7	0.6319	**0.6332** (0.7836)	0.6323 (0.8110)
0.9	**0.6355**	0.6237 (0.8969)	0.6244 (0.8708)

Proposition 3 (Deterministic rare-slice enrichment)**.** *Assume the rarity-dominates-penalty condition in Proposition 2 holds for class c and the λ used by the selector. If the class-aware re-ranking step allocates B_c source samples to class c, then* ***AttriSelect*** *includes at least*

$$\min\{B_c, |T_c|\} \tag{17}$$

target-slice examples from that class. A random or stratified selector with the same class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c} \tag{18}$$

target examples in expectation.

Proof. At a fixed greedy step, the class-aware term adds the same bonus $C_c^{(k)}$ to every remaining candidate from class c. Therefore it can change how many samples are selected from each class, but it cannot reverse the within-class ordering induced by the base attribute score $\bar{s}_i$. By Proposition 2, all target-slice examples in class c appear before all non-target examples in the class queue. Selecting B_c samples from that queue must therefore include all target examples when $|T_c| \le B_c$, and B_c target examples when $|T_c| > B_c$. This proves the deterministic bound. For random or stratified sampling under the same class allocation, each selected class-c source example has probability $|T_c|/N_c$ of belonging to the target slice, giving the expectation by linearity. □

The enrichment gap is strict whenever $0 < |T_c| < N_c$ and $B_c > 0$, because

$$\min\{B_c, |T_c|\} \ge B_c \frac{|T_c|}{N_c}. \tag{19}$$

Thus **AttriSelect** provides a semantic coverage guarantee for the repair surrogate, while random, stratified, uncertainty, Diversity, CoreSet, BADGE, PPAL, and Uncertainty Herding require additional alignment between their selection signal and the target slice.

Proposition 4 (Uncertainty can miss an attribute slice)**.** *Consider uncertainty-based selection within each class, with uncertainty score Q_i and class allocation B_c. If the B_c largest uncertainty scores among non-target examples in class c are all larger than every uncertainty score in T_c, then uncertainty selection chooses zero examples from T_c.*

Proof. The claim is deterministic. If at least B_c non-target examples have uncertainty larger than every target example, the top-B_c uncertainty set contains only non-target examples. The same argument applies to any two-stage uncertainty-first method when the first-stage pool

Table 29: ImageNet10 source-only one-shot data selection. Metric: five-seed mean test accuracy (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	Attribute rarity	Semantic CoreSet	Ours
0.1	43.23	41.85	26.86	38.33	42.62	44.46	44.08	**45.00**
0.3	55.84	57.15	44.64	56.16	58.17	56.54	56.55	**58.32**
0.5	64.77	64.14	58.69	64.87	63.19	65.31	62.56	**66.53**
0.7	69.47	71.63	66.71	68.03	69.57	71.01	67.18	**72.02**
0.9	71.34	71.70	72.88	72.89	72.38	72.48	71.24	**72.92**

Table 30: ImageNet10 source-only iterative active learning. Metric: five-seed mean test accuracy (%); Curve Mean averages ratios 0.1–0.6.

Ratio	Random	Entropy	CoreSet	BADGE	Attribute rarity	Object–attribute entropy	Ours-AL
0.1	43.23	43.23	43.23	43.23	43.23	43.23	43.23
0.2	51.11	**52.26**	50.00	51.27	50.93	50.48	51.33
0.3	55.38	54.85	55.01	56.17	52.94	55.95	**56.96**
0.4	58.71	58.63	59.54	60.46	60.47	60.31	**63.02**
0.5	61.97	63.70	63.92	61.89	61.86	63.88	**65.18**
0.6	66.15	66.70	68.16	67.11	65.78	65.79	**68.80**
Curve Mean	56.09	56.56	56.64	56.69	55.87	56.61	**58.09**

contains no target-slice examples: no second-stage diversity or coverage rule can recover target support that is absent from its candidate pool. □

Finally, suppose the repair utility of a selected source subset is

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{20}$$

meaning target-slice samples have larger marginal value for the discovered repair problem. Then the rare-slice enrichment bound directly implies expected utility at least as large as random or stratified selection under the same class allocation, with a strict gap whenever the enrichment gap is strict. This is not a universal proof that the final trained model must dominate every baseline for every ratio; it is a conditional guarantee that the selector spends a fixed budget on more repair-relevant examples when the VLM metadata and count conditions correctly identify the failing slice.

Soft class-balance behavior The enrichment result above is within-class: the soft class-balance term changes how many examples are drawn from each class, but it does not reverse the attribute ordering inside a class. The following two results summarize the role of C.

Proposition 5 (Hard-balance limit)**.** *When $w = \infty$ and every active class has enough candidates, the round-robin limit selects class counts that differ by at most one. Within each class, candidates are still ordered by the attribute score $\bar{s}_i$.*

Proof. The round-robin rule visits one active class at a time and removes the highest-scoring remaining candidate from that class. After each visit, the class is moved to the back of the queue. Thus no class can receive a second additional sample before every other non-empty class has received one. The selected counts across non-empty classes therefore differ by at most one, except when a class runs out of candidates. Since each class queue is sorted by $\bar{s}_i$, the within-class attribute order is preserved. □

Proposition 6 (Finite-weight balance pressure)**.** *Assume $0 < w < \infty$. At greedy step k, let $A_c^{(k)}$ be the highest remaining base score in class c, and assume all active class-head scores lie in an interval of width*

$$\Delta = \max_c A_c^{(k)} - \min_c A_c^{(k)}. \tag{21}$$

Table 31: Sensitivity to the attribute source. One-Shot Mean averages ratios $0.1/0.3/0.5/0.7/0.9$; AL Curve Mean averages ratios 0.1–0.6.

Attribute source	One-Shot Mean	AL Curve Mean	Final ACC at 0.6
Qwen3-VL	62.96	**58.09**	**68.80**
BLIP	62.18	55.99	65.46
InternVL3	**63.27**	56.33	66.23

If class d is selected at step k while another class c is also active, then

$$n_d^{(k)} - n_c^{(k)} \le \frac{t\Delta}{w}. \tag{22}$$

Equivalently, if an active class c trails class d by more than $t\Delta/w$, then class d cannot be selected before class c at that step.

Proof. Since class d is selected, its adjusted score is at least the adjusted score of class c:

$$A_d^{(k)} + w\frac{t - n_d^{(k)}}{t} \ge A_c^{(k)} + w\frac{t - n_c^{(k)}}{t}. \tag{23}$$

Rearranging gives

$$\frac{w}{t}\left(n_d^{(k)} - n_c^{(k)}\right) \le A_d^{(k)} - A_c^{(k)}. \tag{24}$$

Because both class-head scores lie in an interval of width Δ, we have $A_d^{(k)} - A_c^{(k)} \le \Delta$. Therefore $n_d^{(k)} - n_c^{(k)} \le t\Delta/w$. □

This finite-weight result gives the exact sense in which w controls soft balance. When $w = 0$, the bound disappears and the selector reduces to pure attribute ranking. When $0 < w < \infty$, a class with a large deficit receives enough bonus to overcome any bounded attribute-score advantage from an over-selected class. Increasing w tightens the admissible count gap by the factor $1/w$. In the limiting case $w \to \infty$, the admissible gap goes to zero, matching the round-robin behavior in Proposition 5. Thus finite w does not guarantee exact class quotas, but it guarantees that class imbalance can persist only when the over-selected class has a sufficiently larger attribute score at the current step.

L.1 No-Generation Corner Case

This section discusses the corner case in which no generated repair images are available, i.e., $\mathcal{D}_{\text{gen}} = \emptyset$. This setting removes the main mechanism for adding missing semantic support, but **AttriSelect** still has a useful source-selection interpretation. Since $\tilde{n}_a^{\text{gen}}(v) = 0$, the effective count becomes

$$\hat{n}_a(v) = n_a^{\text{src}}(v), \tag{25}$$

and the base score reduces to

$$\bar{s}_i = \sum_{a \in \mathcal{A}} \left[\left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} - \lambda \frac{n_a^{\text{src}}(v_{i,a})}{\sum_{v \in \mathcal{V}_a} n_a^{\text{src}}(v)} \right]. \tag{26}$$

Thus, when $\mathcal{D}_{\text{gen}} = \emptyset$, the frequency penalty is no longer correcting a synthetic flood. It becomes a source-distribution head-value penalty: values that are already common in the source pool receive lower priority. The rarity term still prefers values that are rare within a class, and the soft class-balance term still controls label-level collapse.

Proposition 7 (Source-only rare-slice enrichment)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and let $T \subseteq \mathcal{D}_{\text{src}}$ be a target failure slice with nonzero source support. Suppose the rarity-dominates-penalty condition in Proposition 2 holds for T under the source-only score above, with $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the*

*class-aware re-ranking step selects B_c samples from class c, then **AttriSelect** returns at least*

$$\sum_c \min\{B_c, |T_c|\} \tag{27}$$

source examples from T. A random selector with the same per-class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T|\big] = \sum_c B_c \frac{|T_c|}{N_c} \tag{28}$$

target-slice examples in expectation.

Proof. The proof is identical to the rare-slice enrichment proof in Appendix L, because that argument uses only the within-class ordering induced by $\bar{s}_i$. When $\mathcal{D}_{\text{gen}} = \emptyset$, the score changes only through the replacement $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the source-only score satisfies the rarity-dominates-penalty condition, then all target-slice samples appear before non-target samples in each class queue. Selecting B_c examples from class c therefore includes $\min\{B_c, |T_c|\}$ target examples. The random expectation follows by linearity under sampling from each class with allocation B_c. □

This proposition shows that generation is not required for the selector to improve target-slice coverage when the source pool already contains target examples. In this regime, the method becomes semantic active subset selection over the source pool. It can increase the number of existing rare failure cases in the retraining subset, but it cannot create support for a target slice that is absent from the source pool.

Proposition 8 (Impossibility for absent source slices)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and suppose $T \cap \mathcal{D}_{\text{src}} = \emptyset$. Then every source-only selector $S \subseteq \mathcal{D}_{\text{src}}$ satisfies*

$$|S \cap T| = 0. \tag{29}$$

Consequently, no active-selection rule over $\mathcal{D}_{\text{src}}$ alone can provide target-slice training support.

Proof. Since $S \subseteq \mathcal{D}_{\text{src}}$ and $T \cap \mathcal{D}_{\text{src}} = \emptyset$, we have $S \cap T \subseteq \mathcal{D}_{\text{src}} \cap T = \emptyset$. Hence $|S \cap T| = 0$ for every possible selector. □

The no-generation case therefore separates two roles of **AttriSelect**. If T already appears in the source pool, attribute selection can enrich those existing target examples under the same rarity-dominates-penalty condition as the main result. If T is absent from the source pool, selection alone is insufficient; repair requires either new data collection or generated images that add target-slice support. This also clarifies the role of λ: with no generated images, a large frequency penalty may suppress globally common values even when they are rare within a specific class. In source-only repair, λ should therefore be interpreted as a head-value suppression weight and may need to be smaller than in settings with heavy generated-data injection.

M Hyperparameter Details

Table 32 reports the **AttriSelect** hyperparameters used in the main experiments. The parameters are the rarity exponent α, frequency-penalty weight λ, and soft class-balancing weight w, matching the notation in Eq. 5.

The larger frequency-penalty weight used for KITTI is consistent with the relative size of the generated repair set. All four settings add 1k generated images, but KITTI has only 5,000 source training images, compared with 10,400 for ImageNet10, 20,000 for COCO, and 80,000 for CelebA. The generated images therefore form a larger fraction of the KITTI repair candidate pool, making a stronger penalty on already-supplied attribute values useful for preserving source-data complementarity. This dataset-

Table 32: **AttriSelect** hyperparameters used in the main experiments.

Dataset	α	λ	w
ImageNet10	0.03	0.5	15
CelebA	0.03	0.5	15
COCO	0.03	0.5	15
KITTI	0.01	5.0	15

dependent shift supports the design of the score: the frequency-penalty term adapts the selector to the amount of generated repair data relative to the available source pool.

Table 33 gives the KITTI sensitivity across source-selection ratios. Across several **AttriSelect** parameter settings, the test mAP50–95 changes only modestly, suggesting that the selector is not brittle to small changes in α, λ, and w once the generated-data frequency penalty is present.

Table 33: KITTI **AttriSelect** sensitivity. Metric: test mAP50–95.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.01	5.0	15	**0.5372**	**0.5646**	**0.5773**	0.5780	**0.5850**
0.03	0.5	15	0.5358	0.5637	0.5764	0.5774	0.5842
0.03	5.0	8	0.5366	0.5641	0.5769	**0.5784**	0.5847
0.10	5.0	15	0.5349	0.5640	0.5768	0.5776	0.5839
0.80	10.0	4	0.5337	0.5629	0.5756	0.5769	0.5828

Table 34 shows the same pattern on COCO pose-estimation repair. The selected main setting remains strongest for most ratios, while nearby **AttriSelect** parameter settings stay within a narrow AP band.

Table 34: COCO **AttriSelect** sensitivity. Metric: COCO keypoint AP.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.03	0.5	15	**0.6183**	**0.6306**	**0.6317**	0.6319	**0.6355**
0.01	0.5	15	0.6152	0.6298	0.6309	0.6308	0.6337
0.03	0.5	8	0.6165	0.6289	0.6305	**0.6323**	0.6344
0.03	0.5	0	0.6148	0.6276	0.6292	0.6306	0.6328
0.00	0.0	15	0.6042	0.6288	0.6203	0.6313	0.6268

N COCO Pose-Estimation Attribute-Score Component Ablation

Table 35 reports the corresponding component ablation for COCO pose-estimation repair. The full $R + P + C$ setting is strongest at four of the five ratios and remains close to the best result at the remaining ratio, suggesting that the three terms are complementary in this setting as well.

The leave-one-component-out comparisons show that removing any term reduces average AP relative to the full score, with average drops ranging from 0.00708 to 0.01428. Single-component variants are also less reliable than the full score across ratios. Overall, the COCO ablation supports the same conclusion as ImageNet10: the best repair behavior comes from combining rarity, generated-data frequency control, and soft class balancing rather than relying on any one term alone.

O Selection t-SNE Visualizations

Before measuring selected-set overlap directly, we first visualize the selected subsets in a shared low-dimensional embedding space. These plots are qualitative: the t-SNE coordinates [36] are computed from lightweight image proxy embeddings and are used only for visualization, not for the selection-overlap metric below. Gray points denote source candidates, and colored points denote the subset selected by each method.

The ratio-0.10 plots in Figures 5 and 6 show the selected subsets for COCO and KITTI. The ratio-0.05 KITTI plots in Figures 7 and 8 provide a more stringent small-budget view, with the overlay figure making the separation between **AttriSelect** and representative active-learning baselines easier to inspect.

Table 35: COCO pose-estimation attribute-score component ablation. Metric: COCO keypoint AP.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$R+P+C$
0.1	0.6048	0.6099	0.5983	0.6067	0.6066	0.6149	0.5954	**0.6183**
0.3	0.6040	0.6134	0.6038	0.6136	0.6142	0.6196	0.6136	**0.6306**
0.5	0.6209	0.6074	0.6113	0.6149	0.6295	0.6246	0.6155	**0.6317**
0.7	0.6151	0.6222	0.6231	**0.6319**	0.6317	0.6302	0.6282	0.6311
0.9	0.6091	0.6175	0.6174	0.6270	0.6276	0.6225	0.6231	**0.6355**

The visual pattern should be interpreted in light of the selection score in Section 4.3. **AttriSelect** is not designed to be a uniformly spread coreset in the t-SNE plane. It behaves more like an attribute-conditioned real-source search: the selected images are related to the repair attributes, but they are spread across more diverse backgrounds, classes, and nearby contexts. The rarity term $R_{i,a}$ intentionally gives higher priority to source examples whose attribute values are rare within their class, so **AttriSelect** can place more mass on peripheral or less common regions of the proxy embedding. These peripheral selections are consistent with the intended rare-case behavior, although the t-SNE plot alone should be read only as a qualitative diagnostic. The penalty term $P_{i,a}$ reduces the priority of attribute values already supplied by the generated pool, so the selected source points are encouraged to complement the synthetic repair data instead of repeating it. Finally, the soft class-balance correction $C_c^{(k)}$ pulls the greedy selection away from a single high-scoring class or local region. Thus the expected signature is controlled non-uniformity: **AttriSelect** may put more mass on visually or semantically unusual regions than random, uncertainty, or pure coverage baselines, while still retaining cross-class coverage through the balance term.

The KITTI overlay illustrates this qualitative behavior. The **AttriSelect** points do not simply trace the same support as the representative active-learning baselines; they are shifted toward a different subset of the proxy embedding space, including some less central regions. This is consistent with the score's purpose: select real source images that are attribute-relevant and sufficiently diverse to support retraining, with extra weight on rare repair-relevant cases. Because t-SNE is only a two-dimensional proxy visualization, this observation is not a standalone performance claim. It motivates the direct selected-set overlap analysis in Section P, where the same conclusion is tested through Jaccard overlap between selected subsets.

These visualizations should not be interpreted as proof that low overlap improves downstream repair. They instead provide a sanity check that **AttriSelect** is not visibly collapsing onto the same region as the standard active-learning selectors. The quantitative version of this diagnostic is the Jaccard analysis in Section P.

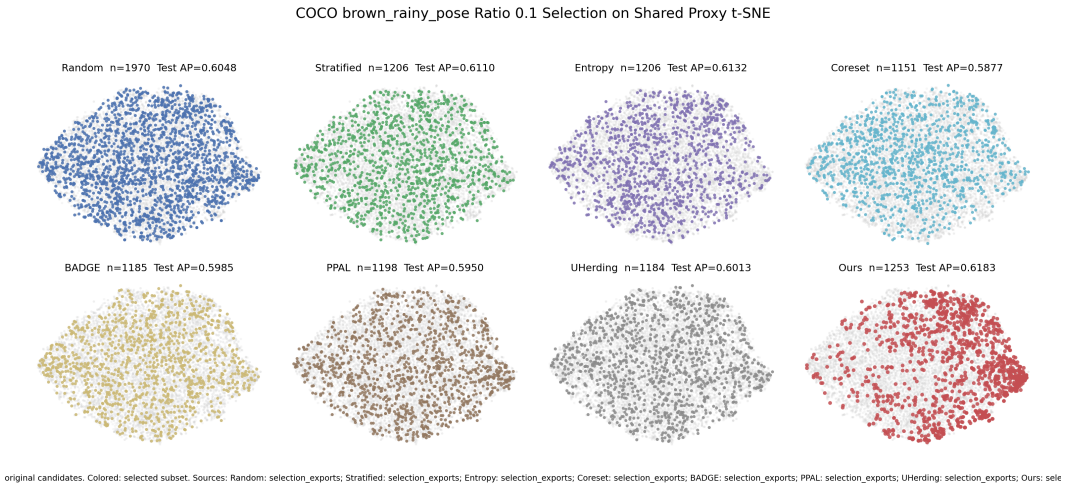


Figure 5: COCO pose-estimation ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

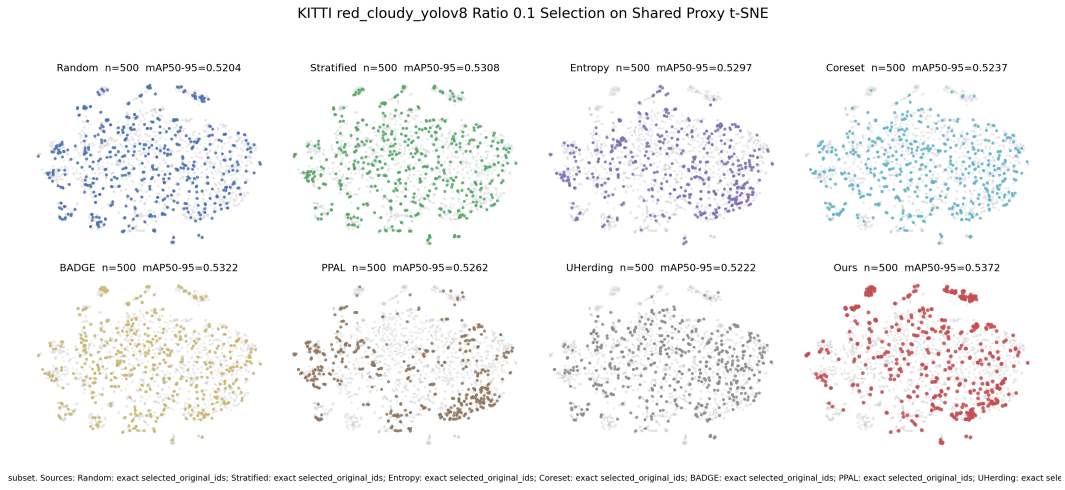


Figure 6: KITTI object-detection ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

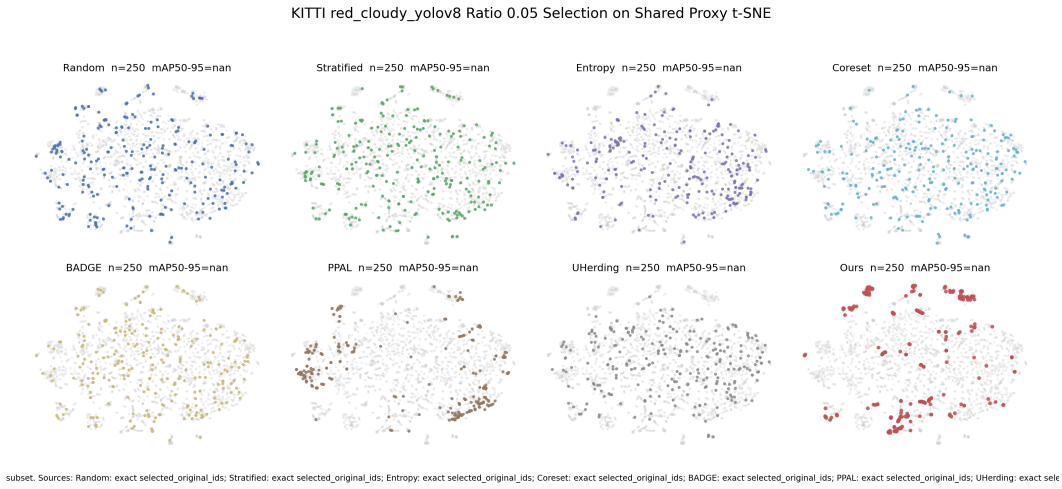


Figure 7: KITTI object-detection ratio-0.05 selected subsets visualized on a shared source-image t-SNE embedding.

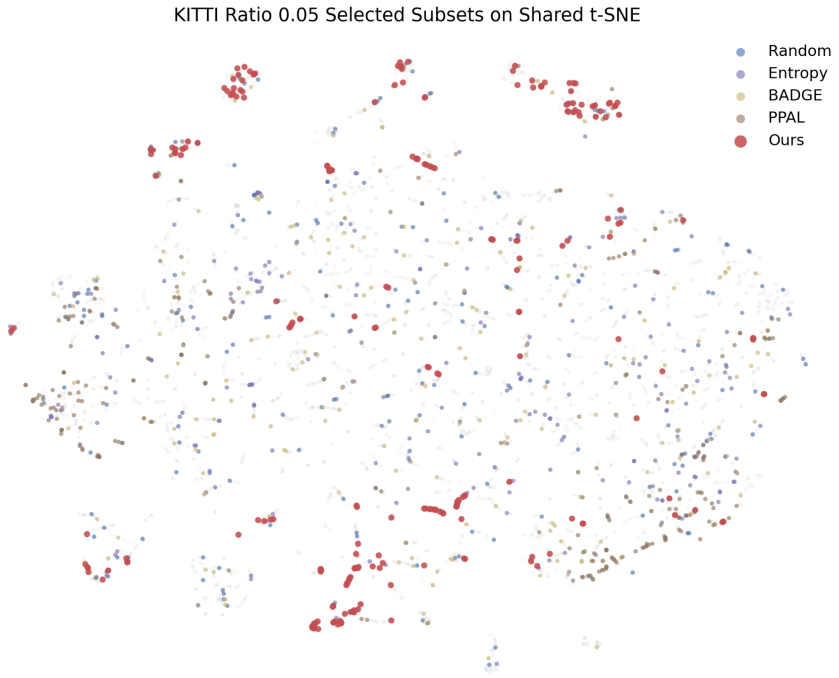


Figure 8: KITTI object-detection ratio-0.05 overlay view for selected active-learning baselines and **AttriSelect**.

P Selection Overlap as an Attribute-Information Diagnostic

This section formalizes what the selection-overlap plots test. Standard active learning selectors rank or cover examples using information such as logits, losses, uncertainty scores, feature embeddings, gradient embeddings, image distances, and sometimes class labels. Let H_i denote all such non-attribute information for source example x_i. Our selector additionally observes structured attribute metadata

$$A_i = \big(a(x_i) : a \in \mathcal{A}\big). \tag{30}$$

Thus the comparison is not only between two scoring formulas. It is a comparison between selectors that are measurable with respect to different information fields: generic AL is attribute-agnostic when it is a function of H_i, while the proposed selector is a function of both H_i and A_i through rarity, generated-data frequency, and class-aware re-ranking.

The overlap plots therefore answer a specific question: is **AttriSelect** merely reproducing the same subset chosen by uncertainty, CoreSet, BADGE, PPAL, Uncertainty Herding, or random selection? Low overlap by itself is not a performance guarantee. However, when low overlap is paired with the slice-enrichment result in Proposition 2 and with improved downstream repair, it supports a stronger interpretation: the gain comes from a semantic selection direction that is not already captured by generic AL signals.

Proposition 9 (Random-overlap scale for attribute-agnostic selectors)**.** *Let the source pool be partitioned into groups $\mathcal{G}_1, \dots, \mathcal{G}_K$, such as classes or class-allocation bins, with $N_g = |\mathcal{G}_g|$. Let S_A be an attribute-driven selected set and let S_H be an attribute-agnostic selected set. Suppose S_A selects $m_{A,g}$ examples from group g and S_H selects $m_{H,g}$ examples from group g. If, conditional on the group counts and on S_A, the attribute-agnostic selector is exchangeable with respect to membership in S_A inside each group, then*

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = \sum_{g=1}^{K} \frac{m_{A,g} m_{H,g}}{N_g}. \tag{31}$$

In the ungrouped case, if both selectors choose $m = rN$ examples from a pool of size N, then

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = r^2 N. \tag{32}$$

Moreover, as $N \to \infty$ with fixed ratio r, the Jaccard overlap

$$J(S_A, S_H) = \frac{|S_A \cap S_H|}{|S_A \cup S_H|} \tag{33}$$

converges in probability to the random-subset scale

$$J_{\mathrm{rand}}(r) = \frac{r}{2 - r}. \tag{34}$$

Proof. Fix a group $\mathcal{G}_g$ and condition on S_A. Since S_H is exchangeable with respect to membership in S_A within the group, selecting $m_{H,g}$ examples from $\mathcal{G}_g$ gives a hypergeometric intersection with mean

$$m_{H,g} \frac{m_{A,g}}{N_g}. \tag{35}$$

Summing over groups gives the first claim. In the ungrouped case this reduces to $m^2/N = r^2 N$. Hypergeometric concentration gives $|S_A \cap S_H|/N \to r^2$ in probability. Since $|S_A \cup S_H| = 2rN - |S_A \cap S_H|$, the continuous mapping theorem gives

$$\frac{|S_A \cap S_H|}{|S_A \cup S_H|} \to \frac{r^2}{2r - r^2} = \frac{r}{2 - r}. \tag{36}$$

□

Proposition 9 gives a useful diagnostic. If an attribute-aware selector and an attribute-agnostic AL selector have Jaccard overlap close to $J_{\mathrm{rand}}(r)$, then the two selected sets are no more similar than

two independent budget-matched subsets at that ratio. This is exactly what one expects when the attributes carry selection information not present in the AL variables H_i. Conversely, if a conventional AL method were already aligned with the discovered attribute slice, its overlap with the selected set from **AttriSelect** would tend to rise above this random-overlap scale.

The ungrouped value $J_{\mathrm{rand}}(r)$ is a useful lower reference for selectors that share a group allocation over imbalanced groups. Suppose two exchangeable selectors both choose B_g examples from group g, with $\sum_g B_g = rN$ and $\sum_g N_g = N$. Proposition 9 gives expected intersection $\sum_g B_g^2/N_g$, and Cauchy–Schwarz gives

$$\sum_g \frac{B_g^2}{N_g} \geq \frac{\left(\sum_g B_g\right)^2}{\sum_g N_g} = r^2 N, \tag{37}$$

with equality only when B_g/N_g is constant across groups. For fixed total budget, the large-pool Jaccard reference is an increasing function of the intersection size, so class-aware allocations on imbalanced datasets can have a random-overlap scale slightly above the ungrouped limit. This helps explain why the baseline–baseline averages in Table 36 are consistently above $J_{\mathrm{rand}}(r)$ when the baselines share similar class-allocation effects. When two methods use different group allocations, the full grouped expression in Proposition 9 is the appropriate reference.

Proposition 10 (Non-redundant repair utility)**.** *Assume the repair-utility model from Appendix L,*

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{38}$$

*where T is the target failure slice. Suppose **AttriSelect** satisfies the deterministic rare-slice enrichment condition in Proposition 3, while an attribute-agnostic selector satisfies the conditional-independence condition in Proposition 1, under the same per-class allocations B_c. Then the expected utility advantage of **AttriSelect** over the attribute-agnostic selector is at least*

$$(u_T - u_0) \sum_c \left[\min\{B_c, |T_c|\} - B_c \frac{|T_c|}{N_c}\right]. \tag{39}$$

This lower bound is strict whenever at least one class has $0 < |T_c| < N_c$ and receives a positive budget that does not occupy the whole class, i.e., $0 < B_c < N_c$. If, in addition, the empirical Jaccard overlap between the two selected sets is close to $J_{\mathrm{rand}}(r)$, then this utility advantage cannot be explained by selecting the same source examples as the attribute-agnostic baseline.

Proof. Proposition 3 gives at least $\min\{B_c, |T_c|\}$ target examples for **AttriSelect** in class c. Proposition 1 gives $B_c|T_c|/N_c$ target examples in expectation for an attribute-agnostic selector under the same class allocation. Since both selectors use the same total budget, the $u_0|S|$ term cancels. Multiplying the difference in target-slice support by the marginal utility gap $u_T - u_0$ gives the stated bound. The strictness condition follows because $\min\{B_c, |T_c|\} > B_c|T_c|/N_c$ whenever a nontrivial target slice receives budget without occupying the whole class. The final statement follows from Proposition 9: near-random Jaccard overlap means the two selectors are not choosing substantially the same subset beyond what the sampling ratio itself forces. □

Table 36 summarizes the observed Jaccard overlaps between selected subsets. The t-SNE coordinates are not used in the calculation; the metric is computed only from selected-set membership. Across ratios, the average overlap between **AttriSelect** and the baselines is close to the random-subset scale $J_{\mathrm{rand}}(r)$ and lower than the average pairwise overlap among baselines. This supports the interpretation that **AttriSelect** operates along a different semantic information direction rather than imitating a conventional AL heuristic.

The corresponding heatmaps are shown in Figures 9, 10, and 11. The diagonal entries are one by definition; off-diagonal entries measure pairwise Jaccard overlap between selected source subsets.

These numbers and figures should be read together with the propositions above. They do not prove that lower overlap alone improves a model. Rather, they show that **AttriSelect** is not redundant

Table 36: Selection-overlap diagnostic. "Ours–baseline" is the mean Jaccard overlap between **AttriSelect** and the non-attribute baselines. "Baseline–baseline" is the mean pairwise Jaccard overlap among non-attribute baselines. Each row averages over ImageNet, CelebA, COCO, and KITTI.

Selection ratio	$J_{\mathrm{rand}}(r)$	Datasets	Ours–baseline	Baseline–baseline
0.10	0.0526	4	0.0533	0.0554
0.30	0.1765	4	0.1766	0.1828
0.50	0.3333	4	0.3323	0.3416

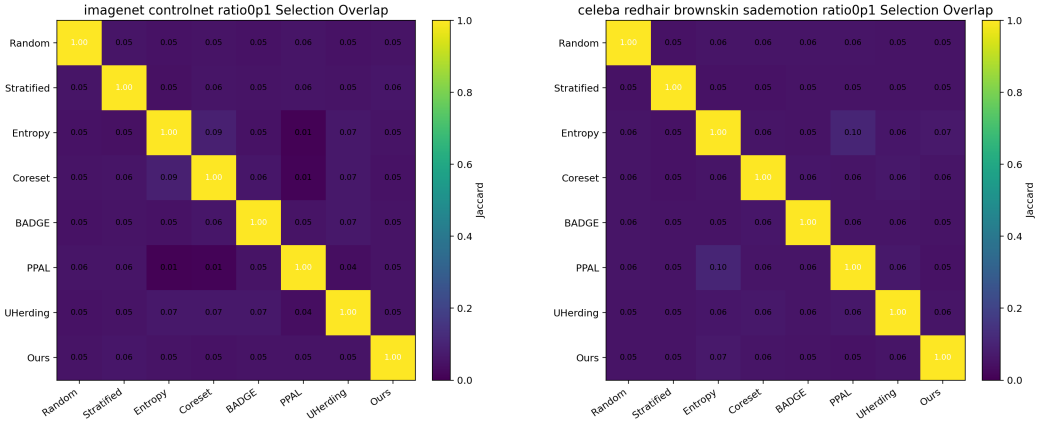
with uncertainty, diversity, geometry, detector-difficulty, or gradient-embedding selectors. The improvement mechanism is therefore consistent with the theory: the selector spends budget on an attribute-defined repair signal that the generic AL variables do not explicitly encode.

Q Limitations and Practical Considerations

VLM dependency.
AttriSelect assumes that source images have VLM-assigned attribute records before selection and relies on accurate VLM annotation.

Attribute vocabulary coverage.
AttriSelect can only target failure modes described by the available attribute vocabulary. If an important failure factor is missing from $\mathcal{A}$ or merged into a coarse closed-set value, the selector cannot explicitly prioritize it.

Generated-data quality.
AttriSelect chooses source examples that complement generated repair images, but it does not correct failures in the generator itself. If generated images miss the target attributes, change labels, or introduce artifacts, the absolute repair gains can decrease even when source selection remains useful.



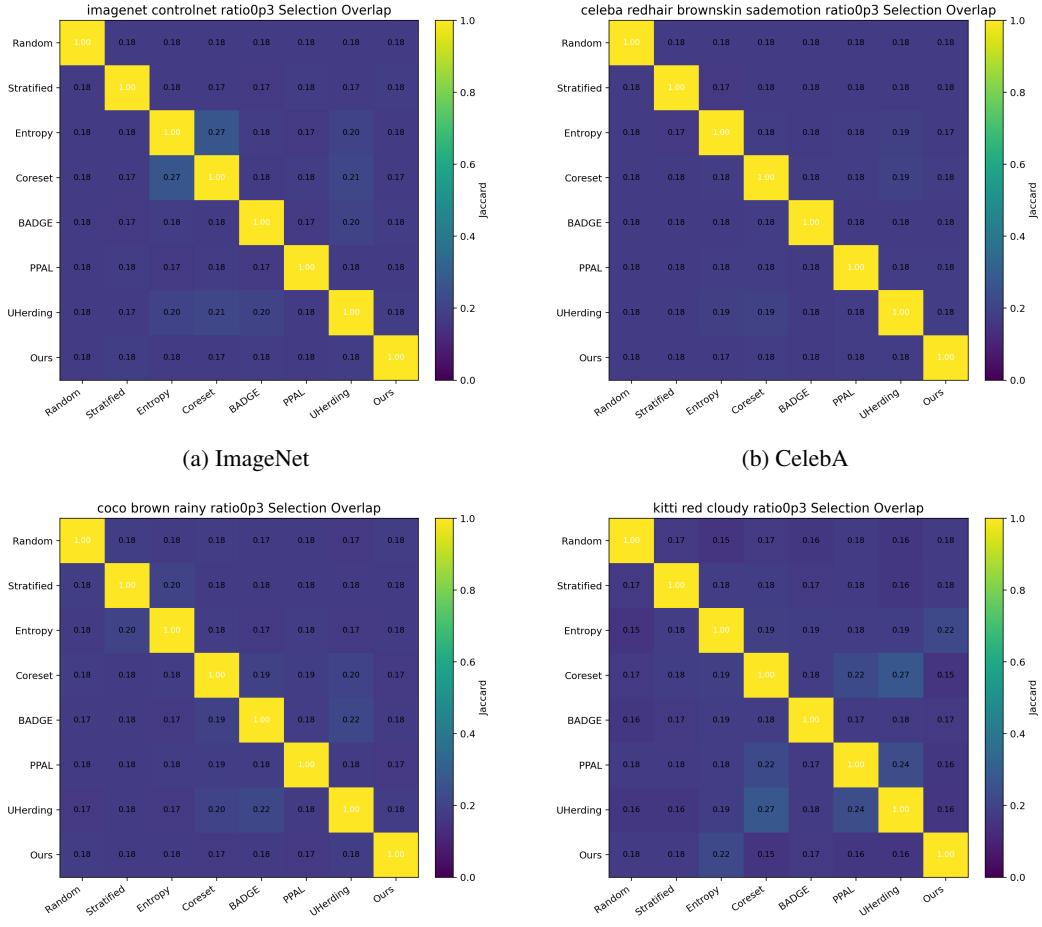
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 9: Pairwise Jaccard overlap among ratio 0.10 selected subsets.



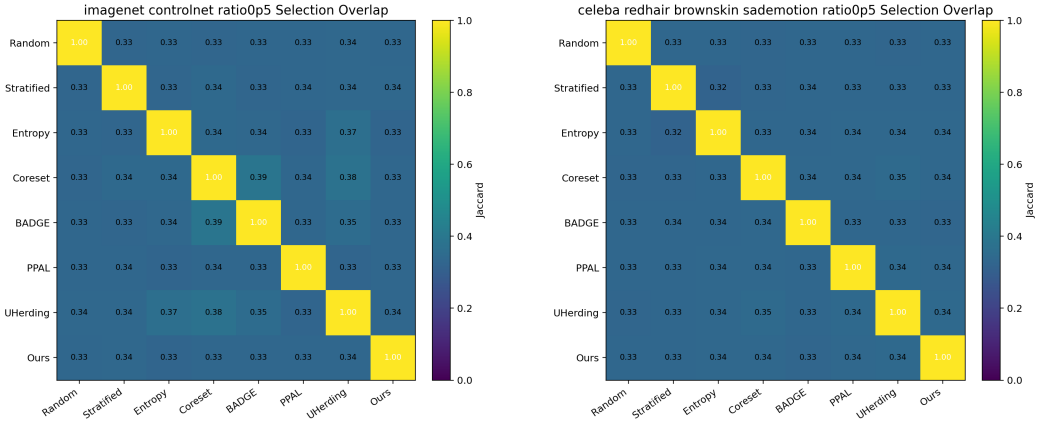
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 10: Pairwise Jaccard overlap among ratio 0.30 selected subsets.



(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 11: Pairwise Jaccard overlap among ratio 0.50 selected subsets.